The Patient Compass: Domain Walls, Oxygen Gating, and the 3-D Survival Manifold of 198,862 Tumors

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- [1] "Active Transport on the Prime Gas: Flat-Band Condensation, the Rabi Phase Transition, and the Arithmetic Qubit," Matos (2026), DOI: 10.5281/zenodo.19243258
- [2] "The Arithmetic Black Hole: Softmax Thermodynamics and the Four Eigenvalue Laws of the Prime Gas," Matos (2026), DOI: 10.5281/zenodo.19442006
- [3] "The Unity Clock: Effective Dimensional Collapse of the Addition-Multiplication Commutator on Coprime Lattices," Matos (2026), DOI: 10.5281/zenodo.19478727

Companion patents:

- [4] "Fault-Injection-Immune Computational Unit Using Primorial Coprime Residue Topology," Matos (2026), U.S. Provisional Patent Application No. 64/031,440 (filed April 7, 2026)
- [5] "Holographic Eigen-Solver Using QM Boundary Projection on Coprime Residue Lattices," Matos (2026), U.S. Provisional Patent Application No. 64/033,689 (filed April 8, 2026)

Abstract

We analyzed somatic mutation data from **198,862 tumors** across **378 independent studies** (cBioPortal), mapping 45 cancer driver genes to the eight hallmarks of cancer (Hanahan & Weinberg, 2000; 2011). We began with a pair completeness hypothesis — grouping the eight hallmarks into four functional pairs and counting how many pairs are simultaneously active — and systematically tested, extended, and ultimately falsified it as a survival framework. What survived this process is a three-part model that is stronger than what we started with.

Domain walls predict survival. Re-parameterizing the 8-bit hallmark vector into domain-wall coordinates — where each hallmark pair is decomposed into *coupled*, *A-only*, and *B-only* states — yields the dominant survival features. The frustrated proliferator (proliferative signaling without invasion) is the single strongest hallmark-derived predictor: HR $= 1.81$ in hypoxic tissue ($p < 10^{-145}$). TP53-coupled growth suppression + death evasion increases hazard by 32% ($p < 10^{-83}$). Hypoxic coupling (simultaneous immortality + metabolism) is massively protective (HR $= 0.40$). These wall features outperform both raw hallmark counts and pair completeness (ΔAIC $= -220$).

Tissue oxygenation gates all hallmark effects. Classifying patients by host tissue oxygen tension reveals that oxygenation is the strongest independent survival predictor in the multivariate Cox model (high-O_2 HR $= 0.59$; low-O_2 HR $= 1.32$). The survival effect of Genome Instability *inverts* by oxygenation: harmful in normoxic tissue ($p = 0.001$), protective in hypoxic tissue ($p = 8.3 \times 10^{-38}$). Forbidden neighbors on the hallmark hypercube — the 94 states never observed in the data — are protective in hypoxic tissue but weakly lethal in normoxia, with the sign flip confirmed across cancer types ($p = 0.020$).

Eight hallmarks collapse to three dimensions. Marchenko-Pastur random matrix analysis finds exactly 3 eigenvalues above the noise threshold in all oxygenation regimes. A 3-composite-score + O_2 model ($C_{\mathrm{CV}} = 0.6101 \pm 0.0055$, 5 parameters) captures 96.3% of the predictive power of the full 25-parameter interaction model, with zero meaningful overfit. The effective cancer survival space is a TP53 axis, an O_2-adaptive axis, and a Genome Instability toggle.

The pair completeness hypothesis — our starting point — ranks 19th of 105 possible hallmark pairings for survival separation. The biological pairs are a human interpretive framework, not a thermodynamic eigenvector. What the data actually support is a 7-parameter energy function combining log-prevalence, forbidden-neighbor topology, hallmark sum, and O_2 interactions, which captures 87.9% of hallmark model discriminability. The model, gene mapping, O_2 classification, and an interactive patient-facing report are freely available.

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1. Introduction

1.1 The Problem

A patient diagnosed with cancer faces a paradox of information. Genomic sequencing generates thousands of data points per tumor, yet the patient's fundamental questions remain unanswered in accessible terms: *What has my cancer learned to do? What has it not learned? Does that pattern mean anything for how I will do?*

The hallmarks of cancer framework (Hanahan & Weinberg, 2000; updated 2011) provides a natural organizing principle. Rather than presenting patients and clinicians with a list of mutated genes, we can ask: *which of the eight core cancer capabilities has this tumor acquired, and which has it not?*

This reframing — from genes to capabilities — is not merely pedagogical. If the *pattern* of capability acquisition carries survival information, then it is clinically actionable: it tells oncologists which patients to monitor more closely, which drug targets may matter more, and which patients may have tumors with exploitable gaps.

1.2 The Pair Hypothesis

The eight hallmarks are not independent. Proliferative signaling and invasion/metastasis represent two distinct growth strategies. Growth suppression evasion and replicative immortality both involve overcoming restraints on cell division. Resisting cell death and immune evasion both involve surviving threats. Genome instability and deregulated metabolism both involve altering the cell's internal machinery.

These functional tensions motivate the organization of the eight hallmarks into four pairs:

Pair	Hallmark A	Hallmark B
1	Proliferative Signaling	Invasion & Metastasis
2	Growth Suppression Evasion	Replicative Immortality
3	Resisting Cell Death	Immune Evasion
4	Genome Instability	Deregulated Metabolism

A pair is **complete** when a tumor has acquired mutations in *both* hallmarks of the pair. We count the number of complete pairs (0 through 4) as an integer summary of hallmark balance.

The theoretical motivation for this specific pairing comes from the spectral structure of hallmark co-occurrence matrices, described in companion papers [1–3], where these four pairs emerge as the natural involution of a symmetry operator on the hallmark space. In this paper, we set aside the spectral theory entirely and ask the purely empirical question: **does pair completeness predict survival?**

1.3 Scope

We analyze every study on cBioPortal with mutation data and at least 50 samples, excluding cell lines and hematologic malignancies (whose hallmark biology differs from solid tumors). The resulting dataset spans 198,862 tumors across 378 studies from dozens of institutions worldwide, representing one of the largest pan-cancer mutation analyses conducted to date.

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2. Methods

2.1 Data Source

All data were obtained from the cBioPortal for Cancer Genomics public API (https://www.cbioportal.org/api), accessed in March 2026. cBioPortal aggregates published cancer genomics datasets with standardized mutation calls and clinical annotations. No patient-identifiable information was accessed or stored.

2.2 Study Selection

We retrieved all studies from cBioPortal and applied the following inclusion criteria:

- **Mutation data available:** Study must have at least one molecular profile of type `MUTATION_EXTENDED`.
- **Minimum sample size:** At least 50 samples.
- **Solid tumors only:** Studies were excluded if the cancer type or study name matched hematologic terms (AML, ALL, CLL, lymphoma, leukemia, myeloid, myeloma, etc.).
- **No cell lines:** Studies with "cellline" or "ccle" in the study identifier were excluded.

378 studies met all criteria. Patient deduplication was performed within each study using patient identifiers to avoid counting multi-sample patients more than once.

2.3 Gene Panel

We selected 45 well-characterized cancer driver genes that collectively cover the eight hallmarks of cancer. The gene-to-hallmark mapping was constructed from established functional annotations in the cancer biology literature:

Proliferative Signaling (Hallmark 0): KRAS, BRAF, NRAS, ERBB2, ERBB3, MAP2K1, FGFR2, NF1, PIK3CA
Growth Suppression Evasion (Hallmark 1): TP53, APC, RB1, CDKN2A, SMAD4, SMAD2, SMAD3, TGFBR2, FBXW7, ACVR2A, NOTCH1, SOX9, TCF7L2, RNF43, AMER1, CTNNB1
Resisting Cell Death (Hallmark 2): PIK3CA, TP53, PTEN, CREBBP, TSC2
Replicative Immortality (Hallmark 3): ATRX
Invasion & Metastasis (Hallmark 4): CTNNB1, CDH1, FAT1, FAT4
Immune Evasion (Hallmark 5): JAK1
Genome Instability (Hallmark 6): MLH1, MSH2, MSH6, PMS2, POLE, BRCA2, ATM, ARID1A, KMT2C, KMT2D, ATRX
Deregulated Metabolism (Hallmark 7): IDH1, GNAS

Some genes (PIK3CA, TP53, CTNNB1, ATRX) map to multiple hallmarks, reflecting their pleiotropic roles.

A hallmark is considered **active** in a tumor if any gene mapped to that hallmark carries a somatic mutation in that tumor's sequencing data.

2.4 Pair Completeness

For each tumor, we compute the 8-bit hallmark activation vector $\mathbf{h} = (h_0, h_1, \dots, h_7) \in \{0,1\}^8$ and count the number of complete pairs:

$$\text{PairCount} = \sum_{(a,b)\in\mathcal{P}} h_a \cdot h_b \tag{1}$$

where $\mathcal{P} = \{(0,4),(1,3),(2,5),(6,7)\}$ is the set of four hallmark pairs. This yields an integer in $\{0,1,2,3,4\}$.

2.5 Symmetry Measure

For each study independently, we construct the hallmark co-occurrence matrix D_h (log-odds ratios of pairwise hallmark co-occurrence) and derive a spectral symmetry measure σ_z for each patient, following the method described in [1]. This continuous measure captures how a tumor's hallmark pattern aligns with the dominant spectral modes of the study's hallmark co-occurrence structure.

For survival analysis, patients within each study are split at the median σ_z value into "high" and "low" symmetry groups.

2.6 Survival Analysis

Overall survival (OS) data were obtained from cBioPortal clinical annotations where available (OS_STATUS and OS_MONTHS fields). A study was included in survival analysis if it had at least 30 patients with survival data and at least 10 death events.

For each eligible study, we performed a Mann-Whitney U test comparing OS between the high and low σ_z groups. The Mann-Whitney test is non-parametric and does not assume proportional hazards or any distributional form, making it robust for the heterogeneous datasets in this analysis.

We report: - The Mann-Whitney z-statistic and two-sided p-value for each study - The median OS difference (Δ) between groups in months - The **direction**: whether higher symmetry is associated with longer or shorter survival

For multivariate survival modeling (§3.19 onward), we used Cox proportional hazards regression with Efron's method for tied event times and an L2 (Ridge) penalty (penalizer = 0.01), implemented in lifelines 0.30. The domain wall Cox model (§3.23) uses 10 wall features + 2 tissue O_2 indicators = 12 covariates, fitted on 48,675 patients with survival data and an assigned O_2 tier. Model comparison uses Harrell's C-index for discrimination and AIC for parsimony. All concordance values are verified by 5-fold cross-validation (§3.29).

2.7 Meta-Analysis

To combine evidence across studies, we applied Stouffer's method in three forms:

1. **Unweighted Stouffer:** $Z = \sum z_i/\sqrt{k}$, where k is the number of studies tested.
2. **Weighted Stouffer:** $Z_w = \sum w_i z_i / \sqrt{\sum w_i^2}$, with weights $w_i = \sqrt{N_i}$.
3. **Direction-agnostic Stouffer:** $Z_{|z|} = \sum |z_i|/\sqrt{k}$. This tests whether pair completeness carries *any* survival information, regardless of direction.

The direction-agnostic test is the most appropriate for the pan-cancer setting, because we hypothesize that pair completeness is informative but do not claim it has a universal direction.

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3. Results

The first five sections describe the raw structure of the data: how 198,862 tumors distribute across hallmark space, which states are forbidden, and what thermodynamic law governs the distribution.

3.1 The Lopsidedness of Cancer

Finding: 85.8% of tumors have zero complete hallmark pairs.

The distribution of pair completeness across all 198,862 tumors is dramatically skewed toward zero:

Complete Pairs	N	%
0	170,622	85.8%
1	22,412	11.3%
2	4,711	2.4%
3	864	0.4%
4	253	0.1%

This is a structural finding about how cancer develops. Despite having eight capabilities to acquire and a theoretical maximum of four complete pairs, the overwhelming majority of tumors are *lopsided* — they acquire capabilities asymmetrically, activating one side of a functional pair without activating the other.

This has clinical implications. A lopsided tumor — one with zero or one complete pairs — has **biological blind spots**: capabilities it has not acquired. These blind spots may represent therapeutic vulnerabilities. A tumor that has acquired proliferative signaling but not invasion capacity, for example, may be locally dangerous but less likely to metastasize, and treatments targeting proliferation may be more effective precisely because the tumor lacks the compensatory capability of its paired hallmark.

3.2 State Space Concentration

Each tumor's hallmark profile is an 8-bit binary vector $\sigma \in \{0, 1\}^8$, yielding $2^8 = 256$ possible states. We treat these states as vertices of an 8-dimensional binary hypercube, where Hamming-adjacent states (differing in exactly one hallmark) are connected by edges. This lattice formulation — analyzing observed versus forbidden vertices, Hamming neighborhoods, energy landscapes, and spectral properties of the adjacency structure — follows the coprime residue lattice methodology developed in the companion papers [1–3], where the same framework revealed universal spectral concentration on algebraic lattices. Here we apply it to the biological lattice of cancer hallmarks.

Only 162 of the 256 possible hallmark states were observed in the data. The distribution is highly concentrated:

Rank	State	Hallmarks Active	N	%
1	01100000	Growth Suppression Evasion, Resisting Cell Death	41,198	20.7%
2	11100000	Proliferative Signaling, Growth Suppression Evasion, Resisting Cell Death	29,939	15.1%
3	10000000	Proliferative Signaling only	12,504	6.3%
4	11100010	Proliferative Signaling, Growth Suppression Evasion, Resisting Cell Death, Genome Instability	11,373	5.7%
5	01100010	Growth Suppression Evasion, Resisting Cell Death, Genome Instability	11,040	5.6%
6	00000010	Genome Instability only	10,421	5.2%

Rank	State	Hallmarks Active	N	%
7	10100000	Proliferative Signaling, Resisting Cell Death	7,609	3.8%
8	01000000	Growth Suppression Evasion only	6,233	3.1%
9	11101010	Proliferative Signaling, Growth Suppression Evasion, Resisting Cell Death, Invasion & Metastasis, Genome Instability	4,875	2.5%
10	01101000	Growth Suppression Evasion, Resisting Cell Death, Replicative Immortality	4,089	2.1%

The top 10 states account for 70.0% of all patients (139,281 / 198,862). Patients in states with N ≥ 20 account for 99.8% of the total.

The most common single state — Growth Suppression Evasion + Resisting Cell Death (01100000) — represents one in five tumors across all cancer types. This pattern is driven by TP53 mutations, which map to both hallmarks and are the most common somatic mutation in human cancer.

Note that none of the top 3 states contain *any* complete pair. The most common state with a complete pair is rank 9 (11101010, with pair 1 complete: Proliferative Signaling + Invasion & Metastasis), representing only 2.5% of patients.

3.3 Hallmark Prevalence Across Cancer Types

The eight hallmarks vary widely in prevalence across the 378 studies:

Hallmark	Mean Prevalence	Median	Range
Growth Suppression Evasion	44.8%	42.6%	0–98.8%
Resisting Cell Death	43.0%	41.3%	0–98.8%
Proliferative Signaling	29.6%	23.8%	0–100%
Genome Instability	25.6%	22.2%	0–97.2%
Invasion & Metastasis	13.0%	9.1%	0–61.5%
Deregulated Metabolism	4.2%	1.8%	0–100%
Replicative Immortality	4.0%	2.2%	0–82.0%

Hallmark	Mean Prevalence	Median	Range
Immune Evasion	1.8%	1.1%	0–42.0%

The extreme asymmetry in hallmark prevalence explains the lopsidedness finding. Growth Suppression Evasion and Resisting Cell Death — both driven heavily by TP53 — are active in nearly half of all tumors. But their paired hallmarks (Replicative Immortality at 4.0% and Immune Evasion at 1.8%) are rarely active. Pairs 2 and 3 are therefore almost never complete, not because tumors avoid completing them, but because the individual hallmarks on one side of the pair are simply rare in mutation data.

This observation is important for interpretation: **the lopsidedness of cancer, as measured here, is partly a property of the gene panel.** A broader panel — including epigenetic, copy number, and expression data — might detect hallmark activations that mutation data alone misses. This is a limitation we address in §4.3.

3.4 Thermodynamic Distribution of Pair Completeness

The pair completeness distribution (§3.1) follows an exponential decay with remarkable precision. Fitting $\ln P(k) = -\beta k + C$ by ordinary least squares gives:

$$\beta = 1.6283, \quad R^2 = 0.9935 \tag{2}$$

The probability of a tumor having k complete pairs is well described by a Boltzmann distribution:

$$P(k) \propto e^{-\beta k} \tag{3}$$

with effective inverse temperature $\beta = 1.63$ and effective temperature $T = 1/\beta = 0.614$.

Each additional complete pair costs approximately 1.63 nats of selective free energy. The quality of the fit ($R^2 > 0.99$) establishes that pair acquisition is not a random combinatorial process — if it were, the distribution would follow a binomial, not an exponential. Instead, the distribution obeys thermodynamic statistics, where each successive pair faces a free-energy barrier of similar magnitude.

This Boltzmann law governs the distribution of 198,862 tumors across cancer types and institutions. It is the single strongest quantitative regularity in our dataset.

3.5 Selection Pressure Against Pair Completion

Of 256 possible hallmark states, 94 (36.7%) are never observed in 198,862 tumors. These "forbidden" states are not uniformly distributed across the phase space — they are concentrated where specific pairs are present.

Forbidden rate by pair-completeness level:

Complete Pairs	Total States	Observed	Forbidden	% Forbidden	% of Patients
0	81	59	22	27.2%	85.8%
1	108	64	44	40.7%	11.3%

Complete Pairs	Total States	Observed	Forbidden	% Forbidden	% of Patients
2	54	29	25	46.3%	2.4%
3	12	9	3	25.0%	0.4%
4	1	1	0	0.0%	0.1%

The forbidden rate increases from 27% at zero pairs to 46% at two pairs, confirming that higher pair completeness faces stronger selection pressure.

More revealing is the pair-specific decomposition. For each pair, we compute the fraction of states that are forbidden when that pair is complete versus when it is not:

Pair-specific forbidden rate differential:

Pair	Description	Δ Forbidden Rate	Role
2	Growth Suppression Evasion + Replicative Immortality	+24.0 pp	**Barrier** — highest thermodynamic cost
3	Resisting Cell Death + Immune Evasion	+9.4 pp	Moderate barrier
1	Proliferative Signaling + Invasion & Metastasis	−1.0 pp	Neutral
4	Genome Instability + Deregulated Metabolism	−13.5 pp	**Enabler** — lowers system energy

Pair 2 is the hardest to achieve: states containing it are 24 percentage points more likely to be forbidden. Biologically, this reflects the mechanistic independence of its components — evading growth suppression (TP53/RB pathway) and achieving replicative immortality (telomerase/ATRX) require two unrelated evolutionary steps.

Pair 4 is an enabler: states containing it are 13.5 percentage points *less* likely to be forbidden. Biologically, this reflects mechanistic coupling — genome instability (e.g., mismatch repair deficiency) directly drives metabolic reprogramming, and vice versa. IDH1 mutations in glioma exemplify this: a single mutation affects both hallmarks simultaneously.

This pair-specific asymmetry extends to the symmetric states — the 16 states where each hallmark pair is either fully complete or fully absent. Six of these 16 states are forbidden. All 6 forbidden symmetric states require the hard pairs (Pair 2 or Pair 3) without the support of enabler pairs. The 10 surviving symmetric states all contain Pair 4 or Pair 1. Symmetric exclusion is driven by pair difficulty, not by symmetry per se.

The pair completeness hypothesis groups the eight hallmarks into four functional pairs. This act builds the Ising model, tests it exhaustively, and follows the evidence to its breaking point.

3.6 Pan-Cancer Survival Stratification

Of 378 studies, 181 had sufficient survival data for analysis (≥30 patients with OS data, ≥10 death events).

29 studies (16.0%) showed statistically significant survival stratification at p < 0.05 (Appendix A). An additional 11 studies reached p < 0.10.

Under the null hypothesis (pair completeness carries no survival information), we would expect approximately 9 studies (5% of 181) to reach p < 0.05 by chance. Observing 29 represents a 3.2-fold enrichment.

The direction-agnostic Stouffer meta-analysis, which tests whether pair completeness carries *any* survival signal across all 181 studies regardless of direction, yields:

$$Z_{|z|} = +15.29, \quad p = 8.9 \times 10^{-53} \tag{4}$$

This is not marginal. Pair completeness carries survival information across the full breadth of solid tumor types in this dataset with overwhelming statistical confidence.

The unweighted Stouffer test, which accounts for direction, yields z = +2.25, p = 0.025, indicating a net positive direction (higher symmetry associated with longer survival) but weaker due to cancellation between cancer types with opposite directions.

3.7 Glioma: Six Independent Replications

The strongest and most consistent signal comes from diffuse glioma, where **six independent cohorts** all reach p < 0.05 in the same direction:

Study	Source	N	Deaths	Δ OS	p-value
Diffuse Glioma	GLASS Consortium	256	192	+29.0 mo	3.4×10^{-7}
Diffuse Glioma	GLASS Consortium, Nature 2019	189	141	+31.5 mo	3.0×10^{-7}
Glioma	MSK, Clin Cancer Res 2019	784	287	+10.6 mo	1.3×10^{-5}
Diffuse Glioma	TCGA GDC, 2025	462	104	+7.7 mo	2.5×10^{-3}
Brain Lower Grade Glioma	TCGA, PanCancer Atlas	462	106	+7.5 mo	3.6×10^{-3}
Brain Lower Grade Glioma	TCGA, Firehose Legacy	262	63	+10.1 mo	4.4×10^{-3}

In every glioma cohort, higher hallmark balance is associated with **longer** survival. The GLASS Consortium cohorts show the most dramatic separation: approximately 30 months of median survival

difference. The MSK cohort, the largest (N=784), shows 10.6 months.

This pattern replicates across three independent institutions (GLASS, MSK, TCGA), three time periods, and both lower-grade and diffuse histologies. To our knowledge, this is the first demonstration that a mutation-derived hallmark balance measure stratifies glioma survival with this degree of consistency.

3.8 The Direction Split

Among the 29 significant studies, the direction of the survival association splits by cancer type:

Higher balance → Longer survival (18 studies): - Diffuse Glioma (6 cohorts): +7.5 to +31.5 months - Melanoma (2 cohorts): +13.8 to +14.9 months - Breast Cancer (2 cohorts): +7.2 to +9.8 months - Prostate Cancer (1 cohort): +5.2 months - Uterine Endometrial (1 cohort): +8.4 months - Uterine Carcinosarcoma (1 cohort): +13.5 months - Thyroid, Anaplastic (1 cohort): +40.6 months - Bladder Urothelial (1 cohort): +2.6 months - TMB/Immunotherapy mixed (1 cohort): +3.0 months - MSK-IMPACT 50K mixed (1 cohort): +0.6 months

Higher balance → Shorter survival (11 studies): - MSK-CHORD mixed (1 cohort, N=20,914): −2.6 months - Colorectal Cancer (2 cohorts): −4.6 to −29.6 months - Ovarian Cancer (2 cohorts): −7.6 to −14.5 months - Pancreatic Cancer (2 cohorts): −7.5 to −9.6 months - Cholangiocarcinoma (1 cohort): −11.9 months - Adrenocortical Carcinoma (1 cohort): −29.8 months - Prostate Adenocarcinoma (1 cohort): −4.5 months - IDH-mutant Glioma (1 cohort): −11.2 months

This directional split is not noise — it is biology. The cancers where balance predicts worse outcomes (colorectal, ovarian, pancreatic) are predominantly gastrointestinal and gynecologic malignancies where activating multiple hallmark pathways simultaneously may reflect a more advanced, treatment-resistant disease state. The cancers where balance predicts better outcomes (glioma, melanoma, breast) may represent tumors where symmetric capability acquisition reflects a more differentiated, less aggressive phenotype — or where the immune system is more effective against tumors that present a broader antigenic surface.

The single exception within the glioma cohorts — the IDH-mutant diffuse glioma study (MSK, $p = 0.036$, $\Delta = -11.2$ months) — is itself biologically informative. IDH mutations fundamentally alter glioma metabolism and epigenetics, creating a distinct disease entity. That the compass direction flips specifically for IDH-mutant gliomas, while being consistent across six other glioma cohorts, suggests the pair measure is capturing real molecular subtype differences.

3.9 Pair-Specific Survival

Each of the four hallmark pairs individually carries survival information. We performed pooled Mann-Whitney U tests comparing median overall survival between patients with complete versus incomplete pairs across all studies with survival data:

Pair	Complete N	Incomplete N	Median OS (Complete)	Median OS (Incomplete)	Δ OS	z	p
1 (Prolif + Invade)	13,462	132,000	22.8 mo	19.2 mo	+3.6	+14.88	4.4×10^{-50}
2 (GrowSup + Immort)	6,482	138,980	22.4 mo	19.4 mo	+2.9	+8.28	1.3×10^{-16}
3 (Death + Immune)	3,075	142,387	22.7 mo	19.5 mo	+3.2	+8.68	4.1×10^{-18}

Pair	Complete N	Incomplete N	Median OS (Complete)	Median OS (Incomplete)	Δ OS	z	p
4 (Genome + Metab)	4,076	141,386	23.5 mo	19.4 mo	+4.1	+11.78	4.7×10^{-32}

All four pairs are independently significant at $p < 10^{-6}$. The survival advantage associated with pair completion ranges from +2.9 months (Pair 2) to +4.1 months (Pair 4).

Notably, Pair 4 — the enabler identified by forbidden state analysis — shows the largest survival differential (+4.1 months), while Pair 2 — the barrier — shows the smallest (+2.9 months). The thermodynamic difficulty of a pair and its survival impact are inversely related: the "easier" pairs carry the stronger survival signal, possibly because they represent more common and therefore more statistically powered configurations.

These pooled results represent the net direction across all cancer types. As established in §3.8, the direction varies by cancer type. The pair-specific results confirm that the survival signal is distributed across all four pairs, consistent with the hallmark pair framework capturing a multi-dimensional biological axis — not an artifact of any single gene or pathway.

3.10 The Cooperative Effect

If pair completions were independent events (a "non-interacting" Boltzmann model), the transition ratio $P(k+1)/P(k)$ would be a constant equal to $e^{-\beta} = 0.196$. The observed ratios deviate systematically:

Transition	Observed Ratio	Predicted (Boltzmann)	Deviation
0 → 1 pair	0.131	0.196	−33.1%
1 → 2 pairs	0.210	0.196	+7.1%
2 → 3 pairs	0.183	0.196	−6.5%
3 → 4 pairs	0.293	0.196	+49.2%

The first transition (0 → 1) faces a *higher* barrier than the Boltzmann average, while the final transition (3 → 4) faces a *lower* barrier. The ratio increases from 0.131 to 0.293 across the four transitions.

This is a cooperative effect: once a tumor has completed one pair, subsequent completions become progressively easier. In condensed matter physics, this pattern is characteristic of nucleation — an initial barrier followed by accelerating growth once a critical nucleus forms. In cancer biology, this suggests that the first pair completion grants the tumor capabilities (e.g., genomic instability via Pair 4) that catalyze the acquisition of additional pairs.

The cooperative effect produces detectable deviations from pure Boltzmann statistics at $k = 0$ (33% suppressed) and $k = 4$ (49% enhanced). Tumors reaching all four complete pairs are 49% more common than a non-interacting model predicts, consistent with the all-or-nothing character of advanced disease: tumors that begin to accumulate pairs may undergo a phase transition-like acceleration of progression.

3.11 The Pair Interaction Matrix

The cooperative effect (§3.10) establishes that pair completions interact, but does not resolve the structure of those interactions. To extract the pair-pair coupling, we model the system as four binary spins $\sigma_i \in \{0, 1\}$ (pair i complete or incomplete) governed by:

$$H = \sum_i \varepsilon_i \sigma_i + \sum_{i<j} J_{ij} \sigma_i \sigma_j \tag{5}$$

where ε_i is the intrinsic field (pair difficulty) and J_{ij} is the spin-spin coupling. The coupling J_{ij} is estimated as the log-odds ratio of co-occurrence: $J_{ij} = \ln\big(P(\sigma_i{=}1, \sigma_j{=}1)/P(\sigma_i{=}1)P(\sigma_j{=}1)\big)$.

The resulting coupling matrix, computed from all 198,862 tumors:

	Pair 1	Pair 2	Pair 3	Pair 4
Pair 1 (Prolif + Invade)	—	+1.05	+1.55	+1.16
Pair 2 (GrowSup + Immort)	+1.05	—	+1.52	+2.29
Pair 3 (Death + Immune)	+1.55	+1.52	—	+1.74
Pair 4 (Genome + Metab)	+1.16	+2.29	+1.74	—

All six couplings are positive (ferromagnetic), meaning that completing any pair increases the probability of completing any other. This confirms the cooperative effect is universal across pair combinations, not driven by a single pair interaction.

The strongest coupling is $J_{24} = +2.29$ — the interaction between Pair 2 (the barrier) and Pair 4 (the enabler). This is the signature of the enabler mechanism: Pair 4 completion most strongly facilitates the pair that is hardest to achieve alone. Without the enabler, Pair 2 is nearly inaccessible (only 8 of 27 possible Pair-2-only microstates are observed — a 70% forbidden rate, the highest of any single-pair macrostate). With Pair 4 present, Pair 2 becomes accessible (7 of 9 Pair-2+4 microstates observed — only 22% forbidden).

The weakest coupling is $J_{12} = +1.05$ — the interaction between Pair 1 (neutral) and Pair 2 (barrier). Pair 1 completion provides the least assistance in overcoming the Pair 2 barrier.

The eigenvalues of J are $\lambda = \{+4.70, -0.69, -1.70, -2.31\}$. The leading eigenvalue is isolated by a gap of 5.40 from the second eigenvalue, indicating that 72% of the interaction variance is captured by a single cooperative mode. The system is not four independent interactions — it is dominated by a single collective tendency toward simultaneous pair completion.

The per-macrostate forbidden rates reveal the full landscape of thermodynamic accessibility across all 16 pair-states:

Macrostate (P1,P2,P3,P4)	k	Microstates	Observed	Forbidden	N
(0,0,0,0)	0	81	59	22 (27%)	170,622
(1,0,0,0)	1	27	19	8 (30%)	13,906
(0,1,0,0)	1	27	8	19 (70%)	4,511
(0,0,1,0)	1	27	16	11 (41%)	2,086

Macrostate (P1,P2,P3,P4)	k	Microstates	Observed	Forbidden	N
(0,0,0,1)	1	27	21	6 (22%)	1,909
(1,1,0,0)	2	9	3	6 (67%)	1,228
(0,1,0,1)	2	9	7	2 (22%)	1,500
(1,0,1,0)	2	9	6	3 (33%)	976
(1,0,0,1)	2	9	6	3 (33%)	698
(0,0,1,1)	2	9	4	5 (56%)	142
(0,1,1,0)	2	9	3	6 (67%)	167
(1,1,1,0)	3	3	1	2 (67%)	359
(1,1,0,1)	3	3	3	0 (0%)	301
(1,0,1,1)	3	3	2	1 (33%)	179
(0,1,1,1)	3	3	3	0 (0%)	25
(1,1,1,1)	4	1	1	0 (0%)	253

The pattern is striking. Macrostates containing Pair 2 *without* Pair 4 have the highest forbidden rates: (0,1,0,0) at 70%, (1,1,0,0) at 67%, (0,1,1,0) at 67%. But macrostates containing Pair 2 *with* Pair 4 drop dramatically: (0,1,0,1) at 22%, (1,1,0,1) at 0%. Pair 4 rescues Pair 2 from inaccessibility.

This interaction matrix provides the complete empirical specification of the tumor's selective landscape at the pair level. The diagonal elements (pair difficulties ε_i) describe the isolated cost of each pair. The off-diagonal elements (J_{ij}) describe how pairs interact. Together, they define a 10-parameter effective Hamiltonian that governs the distribution of 198,862 tumors across $2^4 = 16$ macrostates with five degrees of freedom.

3.12 The IDH1 Bifurcation: A Phase Transition in Glioma

The Ising model predicts that the enabler coupling $J_{24} = +2.29$ should produce its most dramatic effects in cancer types where Pair 4 completion is common. Diffuse glioma provides the critical test: 27.0% of glioma patients have Pair 4 complete (vs. 2.5% globally — a 10.7-fold enrichment), driven almost entirely by IDH1 mutation.

The survival stratification by Pair 4 status in 4,351 glioma patients yields the strongest single signal in the entire dataset:

Group	N (deceased)	Median OS (mo)	z	p
Pair 4 complete	326	51.4	+20.66	7.8×10^{-95}
Pair 4 incomplete	1,546	15.0	—	—
Δ	—	**+36.4**	—	—

Patients with Pair 4 complete (IDH-mutant) live a median **36.4 months longer** than those without. This is the IDH-mutant favorable subclass, recovered purely from the pair framework without using any molecular subtype labels.

The mechanism is visible in the macrostate distribution. The J_{24} coupling drags Pair 2 along: 25.0% of glioma patients have Pair 2 complete (vs. 4.2% globally — a 6.0-fold enrichment). Of the 1,173 glioma patients with Pair 4 complete, 981 (83.6%) also have Pair 2 complete. The dominant glioma

macrostate is (0,1,0,1) — Pair 2 + Pair 4 complete, all others incomplete — containing 937 patients (21.5% of all glioma). This is the IDH-mutant astrocytoma (ATRX + TP53 + IDH1), expressed as a single thermodynamic ground state.

The glioma pair-completeness distribution is **bimodal**:

k	N	%	Phase
0	3,016	69.3%	Paramagnetic (IDH-wildtype)
1	315	7.2%	Boundary (depleted)
2	967	22.2%	Ferromagnetic (IDH-mutant)
3–4	53	1.2%	Saturated

The depletion at $k = 1$ (7.2%, vs. the monotonic decay predicted by a single Boltzmann distribution) is the hallmark of a first-order phase transition: two coexisting phases with a depleted boundary. Glioma tumors are either paramagnetic (k = 0, IDH-wildtype GBM) or ferromagnetic (k = 2, IDH-mutant astrocytoma), with few tumors at the transition point. The Ising model identifies this as a nucleation event: IDH1 mutation simultaneously completes Pair 4 and, via the dominant J_{24} coupling, drags Pair 2 across the accessibility threshold. The tumor jumps from $k = 0$ to $k = 2$ in a single molecular event.

In breast cancer, the same Pair 4 completion produces the **opposite** survival effect (Δ = −18.1 months, z = −2.20, p = 0.028). The enabler is protective in glioma and destructive in breast — confirming that survival direction is determined not by temperature alone, but by which pairs are completed and the tissue-specific biology of those completions.

3.13 Per-Cancer Effective Temperature

The global Boltzmann fit yields $\beta = 1.63$ across 198,862 tumors. Fitting β independently for each cancer type with $≥$500 patients reveals a three-fold range of effective temperatures:

Cancer Type	N	β	T = 1/β	k = 0 (%)
Clear Cell Renal	740	2.94	0.34	96.1%
Soft Tissue Sarcoma	5,474	2.91	0.34	90.3%
Pancreatic	2,862	2.69	0.37	96.5%
Hepatocellular	2,906	2.61	0.38	90.5%
Cholangiocarcinoma	540	2.55	0.39	90.6%
Esophageal	661	2.50	0.40	89.7%
High-Grade Serous Ovarian	1,372	2.32	0.43	97.0%
Lung Adenocarcinoma	5,272	2.18	0.46	83.5%
Breast (targeted)	2,474	2.17	0.46	82.2%
Lung Squamous	1,453	2.12	0.47	85.8%
Bladder	3,007	2.01	0.50	78.8%
Breast (BRCA)	9,763	1.96	0.51	85.1%
Prostate (PRAD)	5,619	1.88	0.53	95.2%
Melanoma	2,627	1.53	0.66	57.0%
Colorectal	4,596	1.42	0.70	80.7%
Diffuse Glioma	**4,351**	**1.32**	**0.76**	**69.3%**
Gastric (STAD)	1,612	1.25	0.80	75.1%

Cancer Type	N	β	T = 1/β	k = 0 (%)
Uterine Endometrial	3,875	1.04	0.96	66.8%

The coldest cancer (clear cell renal, $\beta = 2.94$) has 2.8× higher selective barriers than the hottest (uterine endometrial, $\beta = 1.04$). Glioma ($\beta = 1.32$) is among the hottest — not because glioma tumors are more aggressive, but because the IDH1 pathway provides a low-energy route to pair completion that is unavailable in colder cancers. The high temperature reflects the thermodynamic accessibility of the (0,1,0,1) ground state, not the clinical aggressiveness of the disease.

The correlation between per-cancer β and survival direction is weak (Pearson $r = -0.13$, $p = 0.49$, $n = 30$ cancer types), indicating that temperature alone does not determine whether balance is protective or destructive. The survival direction depends on which specific pairs are completed and the tissue-specific biology of those completions — the full coupling structure J_{ij}, not merely the scalar temperature.

3.14 The Periodic Table of Oncology — Per-Cancer Coupling Matrices

Having measured the global coupling matrix (§3.11) and per-cancer effective temperatures (§3.13), we now compute the full 4×4 coupling matrix J_{ij} independently for each of 37 cancer types with ≥ 200 patients, using Laplace-smoothed log-odds ratios. This yields the complete Ising parameterization per cancer type: four marginal pair probabilities, six couplings, the eigenvalue spectrum, and the three candidate universal constants. The key findings follow.

The three constants are emergent, not intrinsic. The products $\beta \cdot \alpha_c^2$, λ_1/π, and $(1 - f_{\text{forb}}) \cdot \beta$ hold to high precision for the global 198,862-patient mixture (deviations of 0.08%, 5.6%, and 3.2%, respectively), but no individual cancer type satisfies all three simultaneously. Per-cancer $\beta \cdot \alpha_c^2$ ranges from 1.30 (BCC) to 4.51 (clear cell renal); λ_1/π ranges from 0.24 (BCC) to 3.02 (HGSOC); $(1 - f_{\text{forb}}) \cdot \beta$ ranges from 0.17 (uterine endometrioid) to 0.97 (mixed). The constants characterize the *mixture*, not the individual components. In statistical mechanical terms, each cancer type is a thermodynamic subsystem with its own temperature and coupling structure; the apparent constants emerge when these subsystems are observed in aggregate — analogous to how the ideal gas law holds for a gas mixture even when individual molecular species have different interaction potentials.

Coupling matrices are cancer-specific fingerprints. The coupling matrices are not perturbations around a universal shape — they are qualitatively different by cancer type:

Cancer	N	Strongest Coupling	J value	Weakest Coupling	J value	Phase
Breast (BRCA)	9,763	J(2,4)	+2.78	P1–P2	+0.96	Critical
Prostate (PRAD)	5,619	J(3,4)	+2.76	P1–P2	+2.00	Critical
Soft Tissue Sarcoma	5,474	J(1,3)	+3.33	P1–P2	−0.14	Mixed
Lung Adenocarcinoma	5,272	J(3,4)	+1.05	P2–P3	+0.48	Critical

Cancer	N	Strongest Coupling	J value	Weakest Coupling	J value	Phase
Colorectal	4,596	J(3,4)	+2.06	P1–P2	+1.18	Disordered
Diffuse Glioma	4,351	J(1,3)	+3.12	P3–P4	+0.58	Disordered
Uterine Endometrial	3,875	J(2,4)	+1.95	P1–P3	+0.68	Disordered
Bladder	3,007	J(2,3)	+1.38	P3–P4	−0.17	Mixed

In breast cancer, the enabler–barrier coupling J(2,4) dominates, consistent with the global picture. In glioma, the strongest coupling is J(1,3) = +3.12, linking Pair 1 (Proliferation + Invasion) to Pair 3 (Death + Immune) — a fundamentally different topology. Soft tissue sarcoma shows an *antiferromagnetic* coupling between Pair 1 and Pair 2 (J = −0.14), meaning proliferative-invasive and growth-suppression-immortality signatures are mutually exclusive in sarcoma tumors.

Per-pair survival signatures. Instead of a single z-score (balance vs lopsided), we compute a 4-element survival vector $(z_{P1}, z_{P2}, z_{P3}, z_{P4})$ for each cancer type — the z-score for survival when each pair is individually complete vs incomplete:

Cancer	z(P1)	z(P2)	z(P3)	z(P4)	Pattern
Diffuse Glioma	+4.3	+17.1	+2.3	+20.7	All pairs → longer
Breast (BRCA)	−2.7	−1.8	−1.0	−2.2	All pairs → shorter
Bladder	+3.0	+3.0	−1.3	+2.2	P3 reversal
Uterine Endometrial	−3.2	+0.5	−1.7	−0.2	P1 dominant
Melanoma (SKCM)	+1.5	−0.8	+1.1	+3.1	P4 dominant
Soft Tissue	+1.4	+2.7	−0.9	+2.4	P3 reversal
HGSOC	+0.6	+3.5	—	—	P2 dominant

Glioma and breast cancer occupy opposite poles: in glioma, every pair completion predicts longer survival; in breast, every pair completion predicts shorter survival. Several cancers (bladder, soft tissue) show *mixed* signatures where most pairs predict longer survival but Pair 3 (Death + Immune) reverses direction. This 4-vector fingerprint provides more information than the scalar balance z-score and may resolve cases where the overall balance signal is weak.

No coupling feature predicts survival direction. We tested 17 features derived from the per-cancer coupling matrix (including all six J_{ij}, eigenvalues, pair marginals, forbidden fraction, bimodality index) for correlation with the balance survival z-score across 34 cancer types. The strongest Pearson correlation is P(Pair 4) ($r = +0.77, p = 7.3 \times 10^{-8}$), but this result is driven entirely by glioma. Excluding glioma: P(Pair 4) drops to $r = -0.11$, $p = 0.54$. All 17 features become non-significant ($p > 0.27$) when glioma is excluded. The survival direction is not predictable from any single feature of the coupling matrix.

Phase classification. Based on the coupling matrix topology and effective temperature, we classify each cancer into thermodynamic phases:

- **Disordered** ($\beta < 1.5$, all $J_{ij} > 0$): difg, ucec, coadread, stad, egc, coad, mel, bcc, rcc, uec — high temperature, all couplings ferromagnetic, most tumors at $k = 0$.

- **Critical** ($1.5 \leq \beta \leq 2.5$, all $J_{ij} > 0$): brca, luad, skcm, breast, bowel, prostate, bladder, nsclc, mbn, brain, stomach, cesc, hgsoc — intermediate temperature, strong cooperative mode.
- **Ordered** ($\beta > 2.5$, all $J_{ij} > 0$): pancreas, paad, ccrcc, chol — low temperature, very high $k = 0$ fraction ($> 90\%$), pair completion is thermodynamically expensive.
- **Frustrated** (some $J_{ij} < -1$): lusc — competing ferromagnetic and antiferromagnetic interactions.
- **Mixed** (some J_{ij} slightly negative): soft tissue, blca, hcc, hnsc, esca, lung, apad — local antiferromagnetic couplings reflecting mutually exclusive hallmark pathways.

Glioma is the only cancer with unambiguous first-order phase transition behavior (bimodality index 0.18, where values below 0.5 indicate k=1 depletion). Prostate shows a secondary candidate (index 0.45). All other cancers have unimodal k-distributions consistent with thermal equilibrium in a single phase.

The leading eigenvector is nearly uniform. Across 37 cancer types, the leading eigenvector $\mathbf{v}_1$ of the coupling matrix is approximately $\frac{1}{2}(1, 1, 1, 1)$ — all four components within 0.3–0.65 for most types. The system behaves as a single collective mode where all pairs activate together. The notable exception is glioma, where $\mathbf{v}_1 = (0.63, 0.32, 0.63, 0.31)$: Pairs 1 and 3 (Prolif/Invade and Death/Immune) carry double the weight of Pairs 2 and 4 in the dominant mode, reflecting the IDH1-driven pathway that simultaneously activates metabolism and growth suppression while leaving proliferation and immune evasion relatively uncoupled.

3.15 Exhaustive Hypothesis Testing

The open question from §3.14 — what, if anything, predicts survival direction across cancer types? — demands exhaustive testing. We computed 37 candidate features for each of the 34 cancer types with ≥30 deaths and tested each for Pearson and Spearman correlation with the balance survival z-score under three nested robustness conditions: (1) full dataset, (2) excluding glioma, and (3) excluding glioma and breast cancer.

Features tested. The 37 features fall into six categories:

- *Frustration indices* (2): the Pearson anti-correlation $F = -\mathrm{corr}(\mathbf{P}, \mathbf{R})$ between pair marginals and coupling row sums; the cosine-based frustration $1 - \cos(\hat{\mathbf{P}}, \hat{\mathbf{R}})$.
- *Marginal probabilities* (4): $P(\text{Pair } i)$ for $i = 1, \ldots, 4$.
- *Eigenvalue features* (6): λ_1, λ_2, spectral gap, $\lambda_1/|\lambda_2|$, variance fraction, participation ratio.
- *Coupling topology* (7): $J_{13}, J_{24}, J_{34}, \bar{J}, J_{\max}, J_{\text{range}}, J_{13} - J_{24}$ dominance, count of negative couplings.
- *Compound features* (12): $P_4 \cdot \beta$, P_4/β, effective field $\mathbf{P} \cdot \mathbf{R}$, $\det(J)$, $\mathrm{tr}(J^2)$, population-eigenvector alignment $\cos(\hat{\mathbf{P}}, \hat{\mathbf{v}}_1)$, $P_{\max}/P_{\min}$, pair entropy, ground state energy, Pair 4 share, rank inversion, enabler/barrier ratio.
- *PCA modes* (4): projections of the 6-dimensional coupling vector onto the first four principal components.

Results: what failed.

The frustration index $F = -\mathrm{corr}(\mathbf{P}, \mathbf{R})$ — which would predict that cancers with misaligned marginals and couplings have stronger survival signals — showed no significant correlation with any outcome: $r = +0.18$ with $|z_{\text{balance}}|$ ($p = 0.24$), $r = +0.13$ with bimodality ($p = 0.46$), $r = -0.07$ with λ_2 ($p = 0.70$). All three predictions failed. Excluding glioma made all correlations weaker. The frustration index is dead.

The marginal P(Pair 4) showed $r = +0.77$ ($p = 8.6 \times 10^{-8}$) with z_{balance} in the full dataset — an apparently powerful result. However, the Spearman rank correlation was $\rho = +0.06$ ($p = 0.73$), revealing that the Pearson value is driven entirely by the glioma outlier (P(Pair 4) = 27% vs. median 1.4%). Excluding glioma: $r = -0.11$, $p = 0.54$. Binary threshold analysis at every cutoff from 1% to 10% showed no significant split without glioma (all $p > 0.29$). P(Pair 4) as a continuous phase coordinate is dead.

Temperature (β), all six coupling constants (J_{ij}), all eigenvalue features (λ_1, λ_2, spectral gap), determinant, trace of J^2, ground state energy, PCA modes PC1–PC4, and all compound features (torque, rank inversion, enabler/barrier ratio, effective field) — all showed $p > 0.10$ in every robustness tier. Twenty-seven features are dead across all three conditions.

Results: what survived.

One feature passed Pearson significance ($p < 0.05$) in all three robustness conditions:

$$\cos(\hat{\mathbf{P}}, \hat{\mathbf{v}}_1) \quad \text{vs.} \quad z_{\text{balance}}: \quad r = -0.38\ (p = 0.027), \quad -0.37\ (p = 0.032), \quad -0.44\ (p = 0.012) \tag{6}$$

where $\hat{\mathbf{P}} = \mathbf{P}/\|\mathbf{P}\|$ is the unit population-marginal vector and $\hat{\mathbf{v}}_1$ is the unit dominant eigenvector of the coupling matrix J. The effect is negative: greater alignment of the population state with the coupling ground state predicts *worse* survival from balance.

Physical interpretation. The eigenvector $\mathbf{v}_1$ defines the "easy axis" of the coupling landscape — the direction of minimum energy in the Ising Hamiltonian. When the population's pair-probability vector $\mathbf{P}$ is aligned with $\mathbf{v}_1$, the population sits in a deep potential well. Perturbation toward balance ($k = 2$) costs energy: it moves the system uphill, and shorter survival follows. When $\mathbf{P}$ is misaligned (population displaced from the minimum), balance can *lower* energy by moving toward a more coupled configuration, and longer survival follows.

In concrete terms: soft tissue sarcoma ($\cos\theta = 0.32$, $z = +3.07$) and breast carcinoma ($\cos\theta = 0.42$, $z = +3.64$) are maximally misaligned and show the strongest positive balance effect. Clear cell renal ($\cos\theta = 0.94$, $z = -0.96$) and prostate ($\cos\theta = 0.90$, $z = -1.30$) are maximally aligned with their coupling ground state, and balance hurts.

Robustness characterization:

- *Leave-one-out:* all 34/34 iterations yield $p < 0.05$ (r range: -0.35 to -0.41, $\sigma = 0.012$). No single cancer type drives the effect.
- *Partial correlation:* controlling for $\log(N)$ and β jointly: $r = -0.39$ ($p = 0.024$). The effect is not confounded by sample size or temperature.
- *Large-sample cancers only* ($N > 1{,}000$, $n = 23$): $r = -0.46$ ($p = 0.028$), Spearman $\rho = -0.45$ ($p = 0.030$). Both Pearson and Spearman agree in well-powered types.
- *Bootstrap 95% CI:* $[-0.59, -0.10]$ (full); $[-0.63, +0.01]$ (excluding glioma). The full-dataset CI excludes zero; the glioma-excluded CI barely includes it.

Limitations. The effect explains $R^2 = 0.14$ to 0.19 of variance (14–19%). With 37 features tested, it does not survive Bonferroni correction (threshold $p < 0.0014$; observed $p = 0.027$) or Benjamini-Hochberg FDR at $q = 0.05$. Spearman correlation is marginal in the full dataset ($\rho = -0.30, p = 0.083$) though it reaches significance in $N > 1{,}000$ types ($\rho = -0.45, p = 0.030$). The population-eigenvector alignment is the best available scalar predictor but is not definitive.

The pair entropy confound. The cosine alignment $\cos(\hat{\mathbf{P}}, \hat{\mathbf{v}}_1)$ correlates at $r = 0.95$ with the Shannon entropy of the pair-probability vector. This means the alignment effect is largely equivalent to: cancer types with low pair-entropy (one pair dominates) tend to benefit from balance; cancer types with high pair-entropy (many pairs active) tend not to benefit. The two features are mathematically near-collinear and cannot be distinguished at $n = 34$.

Conclusion. From 37 features, 36 are dead. One is wounded but standing. The data's clearest message is negative: no single scalar parameter governs survival direction across 37 cancer types. The survival direction is an emergent property of the full 4×4 interaction between the population state $\mathbf{P}$ and the coupling landscape J — it cannot be compressed to a scalar without losing 81–86% of the information. The critical point described by the three universal constants ($\beta \cdot \alpha_c^2 = 5/2, \lambda_1/\pi = 3/2, (1 - f_{\text{forb}}) \cdot \beta = 1$) is the centroid of a 37-point cloud in coupling space, not a single state of matter.

3.16 Study-Level Decomposition — Unblending the Mixture

The preceding analyses (§3.13–3.15) aggregate all patients of a given cancer type into a single data point, yielding $n = 34$ cancer types. This averaging may mask within-tissue variation: different studies of the same cancer type (e.g., 11 lung adenocarcinoma cohorts) sample different patient populations, sequencing eras, and treatment regimes. If the control signal hides in within-tissue variance, aggregation would destroy it.

We decomposed the analysis to the study level: for each of the 155 studies with ≥10 deaths and a single dominant cancer type, we computed a study-level pair-probability vector $\mathbf{P}_{\text{study}}$ from that study's patients, then measured the cosine alignment $\cos(\hat{\mathbf{P}}_{\text{study}}, \hat{\mathbf{v}}_1)$ against the cancer-type's dominant coupling eigenvector $\hat{\mathbf{v}}_1$ (from the per-cancer J matrix, §3.14). This decomposition treats J as a tissue property (coupling landscape) and $\mathbf{P}$ as a cohort property (where this particular study's population sits in that landscape).

Result: the signal collapses. Study-level $\cos(\hat{\mathbf{P}}, \hat{\mathbf{v}}_1)$ vs. study-level z_{balance} yields $r = -0.096$ ($p = 0.23$) across 155 studies — not significant. The within-tissue partial correlation, controlling for cancer type via fixed effects ($n = 137$ studies from types with ≥ 3 cohorts), is $r = -0.034$ ($p = 0.70$). No individual cancer type with ≥ 3 studies shows a robust correlation. The lone survivor from §3.15 ($r = -0.38, p = 0.027$ at the cancer-type level) does not survive decomposition to the study level.

Study-level pair entropy ($r = +0.053, p = 0.52$) and study-level mean broken pair count ($r = -0.124$, $p = 0.13$) are similarly non-significant. The wounded survivor from §3.15 is now dead at the study level. The population-eigenvector alignment captured a between-tissue signal that does not replicate within tissues.

3.17 The Survival Hessian — Second-Order Interactions and Model-Free Curvature

The pair-level survival analysis (§3.9) treats each pair independently: does completing Pair i predict survival? But pairs may interact: the survival effect of completing Pair i may depend on whether Pair j is also complete. We measured this interaction by computing the synergy:

$$S_{ij} = z_{\text{both}} - z_i - z_j \tag{7}$$

where z_i is the survival z-score for patients with Pair i complete vs. not (globally), and z_{both} is the z-score for patients with both Pair i and Pair j complete vs. neither complete.

Global second-order tensor. The individual pair z-scores are $z_1 = +5.26$, $z_2 = +6.52$, $z_3 = +0.25$, $z_4 = +11.13$. All six pairwise synergies are negative:

	P1	P2	P3	P4
P1	—	−10.1	−4.5	−14.0
P2	—	—	−6.0	−1.2
P3	—	—	—	−9.2
P4	—	—	—	—

Every entry is negative. The pair interactions are universally *antagonistic*: completing two pairs simultaneously produces a survival benefit that is weaker than the sum of completing each alone. This is consistent with diminishing returns — the first pair completion captures the largest survival gain, and subsequent completions add progressively less, because the most informative biological capability has already been acquired.

The strongest antagonism ($S_{14} = -14.0$) occurs between the neutral pair (Prolif + Invade) and the enabler (Genome + Metab), suggesting that these pairs provide overlapping rather than complementary survival information. The weakest antagonism ($S_{24} = -1.2$) links the barrier to the enabler — the biologically most distant pair, whose joint completion provides the most independent information.

Per-cancer synergy correlates with survival direction. Among cancer types with sufficient data to compute the synergy tensor (11 types for S_{14}, 9 for S_{24}, 6 for S_{34}), the synergy magnitude correlates near-perfectly with the balance survival z-score: $r = -0.996$ (S_{14}), $r = -0.979$ (S_{24}), $r = -0.983$ (S_{34}). These correlations are striking but should be interpreted cautiously: synergy S_{ij} and z_{balance} are both derived from survival comparisons in the same data, so partial mathematical coupling is expected.

From crude synergy to model-free curvature. The crude synergy tensor suggested near-perfect prediction of survival direction ($r = -0.996$), but was computed from synergy differences in only 6–11 cancer types with no standard errors. We replaced it with the survival Hessian: the matrix of second partial derivatives $H_{ij} = \partial^2 S/\partial\sigma_i\partial\sigma_j$, estimated via OLS regression on individual patient survival with interaction terms. This model-free approach does not assume a Hamiltonian — it measures curvature directly from the data with proper standard errors.

Global Hessian. For 60,988 deceased patients, we fitted the full 16-parameter Taylor expansion through fourth order: $S(\sigma) = a + \sum g_i\sigma_i + \sum H_{ij}\sigma_i\sigma_j + \sum T_{ijk}\sigma_i\sigma_j\sigma_k + K_{1234}\sigma_1\sigma_2\sigma_3\sigma_4$. The complete expansion is identifiable because the 16 pair states span 2^4 degrees of freedom. Results by order:

Gradient (first order). Pair 1 (Prolif + Invade) completion adds +4.94 months ($t = +11.4, p < 10^{-29}$). Pair 4 (Genome + Metab) adds +2.90 months ($t = +2.7, p = 0.007$). Pairs 2 and 3 are not individually significant ($|t| < 1$).

Hessian (second order). The dominant element is $H_{24} = +26.1$ months ($t = +14.2, p < 10^{-45}$): patients who complete both the barrier (Pair 2) and the enabler (Pair 4) simultaneously gain 26 months beyond the sum of individual pair effects. The 384 patients with pair state $(0, 1, 0, 1)$ — Pair 2 + Pair 4 only — have mean OS 49.5 months vs. 21.0 months for the ground state $(0, 0, 0, 0)$. This is the IDH-mutant astrocytoma signal (§3.12) expressed in the curvature language. The remaining significant elements are $H_{14} = -5.3$ months ($t = -2.4$) and $H_{34} = -8.3$ months ($t = -2.1$) — both antagonistic.

Third-order tensor. One element is significant: $T_{124} = -11.3$ months ($t = -2.6, p = 0.009$), indicating a three-body interaction among Pairs 1, 2, and 4 that is not captured by pairwise terms alone. The

Ising model's pairwise assumption is violated at third order.

Fourth-order coupling. $K_{1234} = +8.5$ months ($t = +0.5$, $p = 0.62$, not significant). The four-body interaction does not contribute detectably.

The Hessian eigenvalues are $\{-26.8, -3.1, +0.5, +29.5\}$ — a saddle structure with one strongly positive and one strongly negative principal curvature. The energy landscape is neither uniformly convex nor concave; it has a ridge.

Per-cancer Hessians. For 30 cancer types with ≥ 50 deceased patients, we fitted the second-order model ($S \sim \sigma_i + \sigma_i\sigma_j$, 11 parameters) and eigendecomposed each Hessian.

Test: does the Hessian predict survival direction? The total interaction $TI = \sum_{i<j} H_{ij}$ across 27 cancer types with nonzero z_{balance} yields $r = -0.15$ ($p = 0.45$). The crude synergy result of $r = -0.996$ does not survive proper estimation across the full set of cancer types. The apparent near-perfect prediction was a small-sample artifact ($n = 6$–11) amplified by mathematical coupling: both the synergy and z_{balance} are derived from survival stratification by pair state. With proper regression-based Hessian estimation and $n = 27$, the total-interaction signal vanishes.

However, the Hessian reveals two real signals:

1. The specific element H_{34} (Death/Immune × Genome/Metab interaction) correlates with survival direction at $r = -0.52$ ($p = 0.005$). Cancer types where this interaction is more antagonistic tend to benefit from balance. This is the only Hessian element that individually predicts direction.

2. The gradient magnitude $\|\nabla S\|$ — how strongly individual pair completions affect survival within a cancer type — correlates with z_{balance} at $r = +0.49$ ($p = 0.009$). Cancer types where pair completions matter more individually also show stronger balance effects.

3. The mean ratio of survival Hessian to thermodynamic coupling $\langle H_{ij}/J_{ij}\rangle$ correlates with survival direction at $r = -0.40$ ($p = 0.041$). The survival-to-coupling conversion factor — how efficiently the Ising structure translates to clinical outcome — varies systematically across cancer types.

Patient-level composite predictor. For 19,646 deceased patients in cancer types with computed Hessians, we fitted OS $\sim n_{\text{walls}} + d_{\text{forb}} + \sigma^T H \sigma$. The domain wall count is the dominant predictor: $\beta = -4.94$ months per wall ($t = -15.1$, $p < 10^{-50}$). Forbidden boundary distance adds independently: $\beta = +5.81$ months per unit ($t = +8.6$, $p < 10^{-17}$). Hessian curvature exposure ($\sigma^T H \sigma$) is not significant as a patient-level feature ($t = +1.0$, $p = 0.30$). The composite model explains $R^2 = 1.2\%$ of survival variance — small in absolute terms but highly significant, and consistent with the expectation that state-level features capture only a fraction of the variance in a quantity dominated by stage, treatment, and tumor biology.

Verdict. The Hessian approach burned the crude synergy tensor's $r = -0.996$ — that signal was noise, not the organizing principle. What survives: the barrier-enabler synergy $H_{24} = +26.1$ months is a real global phenomenon driven by the IDH-mutant glioma axis; the H_{34} element specifically predicts cross-cancer survival direction; domain wall count and forbidden boundary distance are significant patient-level survival features; and the Ising model's pairwise assumption is violated at third order (T_{124}, $p = 0.009$). The survival landscape has genuine curvature, but that curvature does not collapse to a single scalar predictor of direction. The landscape is a saddle — and the full 4×4 eigenstructure, not any scalar summary, is required to navigate it.

3.18 Simpson's Paradox — Direction Flips at the Aggregation Boundary

The study-level decomposition (§3.16) permits a direct test for Simpson's paradox: does the survival direction *reverse* between the study level and the cancer-type level? For each cancer type with ≥ 2 studies, we compared the cancer-type z_{balance} (all patients aggregated) with the mean of study-level z-scores.

Of 28 cancer types tested, **12 show direction flips (43%)**. Notable examples:

Cancer type	Cancer-level z	Mean study-level z	Flip?
Breast carcinoma	-2.31	$+0.71$	YES
Lung adenocarcinoma	$+1.59$	-0.28	YES
High-grade serous ovarian	$+3.37$	-1.54	YES
Soft tissue sarcoma	$+3.07$	-0.57	YES
Esophageal	-2.40	$+0.89$	YES
Uterine endometrial	-3.15	$+1.35$	YES

In breast cancer, aggregating all patients shows balance is harmful ($z = -2.31$), but the average *individual study* shows balance is beneficial ($z = +0.71$). In soft tissue sarcoma, the aggregated signal is strongly positive ($z = +3.07$), but the average study-level signal is negative ($z = -0.57$).

A 43% direction-flip rate means that for nearly half of cancer types, the relationship between pair completeness and survival depends on *which patients are pooled*. This is a classic Simpson's paradox: a trend present in aggregated data reverses in disaggregated subgroups. The confounding variable is likely the study composition — studies within the same cancer type differ in stage, treatment, sequencing technology, and institutional patient mix, all of which correlate with both pair completeness and survival.

Implications. The Simpson's paradox result has two consequences. First, it explains why scalar predictors fail (§3.15): the cancer-type-level signal is an average over study-level signals that often point in opposite directions. Averaging conflicting signals produces a weak, noisy aggregate. Second, it suggests that the true survival effect of pair completeness is study-specific rather than cancer-type-specific — the relevant covariate for predicting direction is not the tissue of origin but the specific patient population, treatment context, and sequencing cohort. This motivates prospective, study-stratified validation rather than pan-cancer aggregation.

3.19 The Rebinning Offensive — Cox Hazards, State Rebinning, and Pair Discovery

The preceding analyses used OLS regression on deceased patients only, discarding 83,925 censored (alive) observations. They also binned exclusively by cancer type, leaving open the possibility that the hallmark state itself — the 8-bit molecular signature — is the more fundamental unit. Finally, the four biological pairs [(0,4), (1,3), (2,5), (6,7)] were assumed, not discovered. We attacked these three assumptions simultaneously.

Cox Proportional Hazards Hessian (log-hazard space). We refitted the full 2^4-state interaction model as a Cox PH regression on all 145,462 patients with survival data (61,537 events, 83,925 censored). The Cox model outputs log-hazard ratios, which are dimensionally commensurable with the log-odds coupling matrix J — both live in logarithmic probability space. The Cox Hessian is:

$$H^{\text{Cox}} = \begin{pmatrix} 0 & -0.131 & -0.142 & +0.040 \\ -0.131 & 0 & -0.040 & -0.528 \\ -0.142 & -0.040 & 0 & +0.310 \\ +0.040 & -0.528 & +0.310 & 0 \end{pmatrix} \tag{8}$$

The dominant element is $H_{24}^{\text{Cox}} = -0.528$ log-HR ($z = -7.46$, $p < 10^{-13}$): simultaneous completion of the barrier (Pair 2) and enabler (Pair 4) reduces the hazard rate by $1 - e^{-0.528} = 41\%$. This is the Cox analog of the OLS finding $H_{24} = +26.1$ months; the sign reversal reflects the hazard-vs-time inversion (lower hazard = longer survival). The only hazard-increasing interaction is $H_{34}^{\text{Cox}} = +0.310$ ($z = +2.01$, $p = 0.045$): Death/Immune × Genome/Metab completion *increases* hazard by 36%. Third-order terms remain significant ($T_{124} = -0.601$, $z = -2.37$; $T_{124} = +0.486$, $z = +2.88$), confirming the pairwise Ising violation. Overall concordance: $C = 0.5187$ — the pair interaction structure, despite its statistical significance, has essentially zero predictive power at the individual patient level.

Phase space: probability × lethality. Computing the empirical hazard rate (deaths / person-months) and observation probability for all 16 macrostates reveals that pair count k is the single strongest predictor of hazard across macrostates: $r = -0.832$ ($p = 6.4 \times 10^{-5}$). The ground state (0000, 85.8% of patients) has hazard 0.0154 deaths/month; the fully completed state (1111, 0.13% of patients) has hazard 0.0032 — a 4.8× reduction. Probability and lethality are only marginally correlated ($r = +0.45$, $p = 0.078$), suggesting that the thermodynamic fitness landscape (which states exist) partially decouples from the clinical outcome landscape (which states kill).

State vs. cancer type. Among 63,121 deceased patients with cancer type annotations, one-way η^2 decomposes survival variance as: hallmark state (8-bit signature) explains 3.89%, cancer type explains 16.10%, and their interaction explains 25.42%. Cancer type is 4.1× more predictive than hallmark state alone. Furthermore, of 22 states observed in $\geq$ 3 cancer types with $\geq$ 20 patients each, 21 (95%) show statistically significant survival differences across cancer types *within the same state* (all $p < 10^{-9}$). Cancer type is not an illusion. However, hallmark state contributes an incremental 9.33% of variance on top of cancer type, confirming that molecular architecture carries independent information. Both covariates are real; neither is sufficient alone.

Random rebinning null model. A 1,000-permutation test confirms that cancer-type labels explain far more variance than random labels ($\eta^2 = 0.161$ vs. null mean 0.001, $p < 0.001$). State labels similarly survive the null test ($p < 0.001$). Both binning schemes carry genuine signal.

Pair structure discovery. Hierarchical clustering of the 8 × 8 hallmark co-occurrence matrix (Ward linkage, 4 clusters) yields clusters {Immort, Invade, Immune, Genome, Metab}, {GrowSup}, {Death}, {Prolif} — a 5-1-1-1 split that matches only Genome+Metab among the four biological pairs. An exhaustive search over all 105 distinct pairings of 8 hallmarks into 4 pairs, ranked by ANOVA F-statistic for pair count → survival, finds the WINNER_PAIRS at rank 19 ($F = 72.9$). The data-optimal pairing is (Prolif, Immort) | (GrowSup, Death) | (Invade, Immune) | (Genome, Metab), with $F = 188.2$ — 2.6× stronger survival separation. Notably, GrowSup+Death appears in all top-10 pairings, suggesting that these two hallmarks are more tightly coupled to survival than the biological pairing (GrowSup, Immort) reflects. The biological pairs are structurally motivated (Hanahan & Weinberg); the data-optimal pairs are survival-motivated. They are not the same.

Universal constants from pure states. Recomputing the three universal constants from the full patient pool (ignoring cancer type): $\beta \cdot \alpha_c^2 = 2.498$ (target 5/2, deviation 0.08%) and $\lambda_1(J)/\pi = 1.497$ (target 3/2, deviation 0.23%) hold with extraordinary precision. Bootstrap 95% CI for λ_1/π:

[1.474, 1.519]. However, $(1 - f_{\text{forb}}) \cdot \beta = 0.231$ (target 1, deviation 77%) — this third relation holds only as a per-cancer average, not globally, because the global forbidden fraction ($f_{\text{forb}} = 0.858$) is dominated by the 85.8% of patients in the ground state.

K/J diagnostic in proper units. Dividing the Cox Hessian by the coupling matrix element-wise yields the dimensionless thermodynamic efficiency K/J: the lethality cost per unit of evolutionary effort. The mean $K/J = -0.043$ — pair interactions are net protective. The most protective pair is barrier × enabler ($K/J = -0.230$); the only hazard-increasing interaction is immune × enabler ($K/J = +0.178$). The spectral ratio (leading eigenvalue of H^{Cox} / leading eigenvalue of J) is +0.135.

Per-cancer Cox Hessian vs. survival direction. Fitting Cox PH models independently for 34 cancer types and correlating with the balance z-score: total interaction $r = +0.068$ ($p = 0.70$) — dead, consistent with OLS. H_{34} specifically: $r = +0.156$ ($p = 0.38$) — the OLS signal ($r = -0.52$, $p = 0.005$) does not survive the Cox framework. But per-cancer concordance correlates with z_{balance} at $r = +0.557$ ($p = 6.2 \times 10^{-4}$): cancer types where the Cox model fits well are those where pair completion improves survival. The model's explanatory power is itself a predictor of the outcome it explains — a meta-signal that suggests the Ising framework captures a real but heterogeneous biological process.

3.20 The Grating Smasher — Testing the Pair Assumption Itself

The preceding sections parameterize survival using four hallmark pairs inherited from Hanahan and Weinberg's biological framework. Section 3.22 showed that these pairs rank 19th of 105 possible pairings for survival separation. We now remove the pair assumption entirely and ask: what structure does the data actually want?

Individual hallmark Cox model. Fitting a Cox PH model on all 8 hallmarks independently (145,462 patients, 61,537 events) yields concordance $C = 0.569$ — already higher than any pair-based representation. The hallmark ranking by absolute z-statistic is:

Rank	Hallmark	Coefficient	z	Direction
1	Growth Suppression Evasion	+0.399	+37.1	Hazard-increasing
2	Invasion	−0.249	−21.0	Protective
3	Proliferative Signaling	+0.107	+13.0	Hazard-increasing
4	Immune Evasion	−0.366	−12.3	Protective
5	Deregulated Metabolism	−0.251	−12.2	Protective
6	Replicative Immortality	−0.166	−8.1	Protective
7	Genome Instability	−0.034	−3.8	Protective
8	Resisting Cell Death	+0.031	+3.0	Hazard-increasing

All eight hallmarks are significant ($p < 0.003$). The critical finding: GrowSup is by far the dominant survival predictor, followed by Invade and Prolif. Adding all 28 pairwise interactions (36-term model) raises concordance to $C = 0.577$, with GrowSup × Death as the strongest interaction ($z = +13.2$, $p < 10^{-38}$). This interaction reflects the fact that TP53 mutations simultaneously disable both growth suppression and apoptosis — the data recovers the known biology of the single most important tumor suppressor gene.

Core–Periphery decomposition. Spectral analysis of the 8×8 log-odds co-occurrence matrix reveals a single dominant eigenvalue $\lambda_1 = 3.22$ capturing 69.1% of the interaction variance, with a spectral gap of 2.93 to the next eigenvalue. The leading eigenvector loads exclusively on five hallmarks: Immort (−0.54), Immune (−0.51), Metab (−0.46), Genome (−0.37), Invade (−0.32) — the "malignancy core." The remaining three hallmarks (Prolif, GrowSup, Death) have near-zero loadings ($|v| < 0.05$), forming

a loosely coupled periphery. The participation ratio (effective dimensionality) is 1.92 — the hallmark space is essentially two-dimensional.

Testing Core count (5 hallmarks) vs. Periphery count (3 hallmarks) as Cox predictors: Periphery count alone ($C = 0.543$, $z = +35.2$ hazard-increasing) outperforms Core count alone ($C = 0.529$, $z = -29.6$ protective). Combined: $C = 0.558$. With interaction term: $C = 0.560$, where the interaction is strongly negative ($z = -22.3$): having both core and periphery hallmarks is protective relative to having only periphery, consistent with the domain-wall lethality mechanism. Both WINNER_PAIRS ($C = 0.518$) and OPTIMAL_PAIRS ($C = 0.534$) are inferior to the raw Core + Periphery decomposition.

Exhaustive grouping search. Testing every possible partition of 8 hallmarks into 2 groups (127 partitions), 3 groups (966 partitions), and 4 groups (105 pairings): the best groupings by survival F-statistic scale *inversely* with group count:

Groups	Best F	Best partition
2	287.9	(GrowSup, Death) ‖ (rest)
3	**344.7**	(Prolif, Invade, Immune, Genome) ‖ (GrowSup, Death) ‖ (Immort, Metab)
4	188.2	(Prolif, Immort) ‖ (GrowSup, Death) ‖ (Invade, Immune) ‖ (Genome, Metab)

Three groups is optimal ($F = 344.7$, $p = 5.3 \times 10^{-222}$). The winning three-group partition separates: (a) the TP53 bond (GrowSup + Death), (b) the telomere–metabolism axis (Immort + Metab), and (c) everything else. Notably, GrowSup + Death appears as a distinct group in *all* top-10 three-group partitions, and the top-1 two-group split is exactly (GrowSup, Death) vs. the other six. TP53 is the organizing principle.

LASSO feature selection. L1-regularized regression on all 92 features (8 main + 28 pairs + 56 triples) retains 46 non-zero coefficients ($R^2 = 0.017$ on log(OS)). The top two LASSO features are Prolif × Death ($\beta = +0.143$, longer survival) and Prolif × GrowSup × Death ($\beta = -0.136$, shorter survival). Of 46 survivors, 27 are third-order interactions — the pairwise Ising model misses more than half the signal that L1 regularization retains.

Mutual information. Nonparametric MI analysis confirms: total hallmark count (MI = 0.048) carries more survival information than any pair-based count (WINNER MI = 0.020; OPTIMAL MI = 0.024). Periphery count (MI = 0.042) exceeds Core count (MI = 0.029). Among individual hallmarks, Prolif (MI = 0.023) is most informative; Immune (MI = 0.004) least.

Re-piping the constants with optimal pairs. Recomputing the Ising model with the $F = 188.2$ optimal pairing: $\beta = 1.608$, $R^2 = 0.919$ (vs. 0.994 for WINNER). The Boltzmann fit degrades because 61.4% of patients complete Pair 2 (GrowSup, Death) — the TP53 bond is so common that the exponential decay assumption breaks. The coupling matrix becomes asymmetric: $J_{13} = 2.36$ (Prolif+Immort × Invade+Immune) and $J_{14} = 1.84$ dominate, while $J_{12} = 0.22$ (TP53 bond × Invade+Immune) nearly vanishes — the TP53 pair decouples from the rest. The universal constants shift: $\beta \cdot \alpha_c^2 = 2.466$ (deviation 1.4%), $\lambda_1/\pi = 1.350$ (deviation 10.0%), $(1 - f_{\text{forb}}) \cdot \beta = 1.009$ (deviation 0.9%). Two of three constants *worsen*, but the third (the one that failed globally for WINNER pairs) now hits target. The constants are *pair-choice-dependent*, not universal — they characterize the specific mathematical representation, not the underlying biology.

Per-cancer hallmark ranking. Fitting individual Cox models for 35 cancer types: GrowSup is the top predictor in 8 cancers (including breast, prostate, head/neck, renal), Death in 7 (lung, pancreas, prostate, cholangiocarcinoma), Genome in 5, Invade in 5, Prolif in 5. No single hallmark dominates universally. The glioma signal (concordance $C = 0.755$, far above all others) is driven by Metab ($z =$

−28.3), confirming that the IDH1 bifurcation — a metabolic enzyme mutation — is the strongest single-hallmark signal in the dataset.

The definitive concordance scoreboard:

Model	**C-index**	**Δ above random**
Random baseline	0.500	0.000
k_{old} (WINNER pairs)	0.518	+0.018
k_{new} (OPTIMAL pairs)	0.534	+0.034
n_{total} (hallmark count)	0.521	+0.021
n_{core} (5 hallmarks)	0.529	+0.029
$n_{\text{core}} + n_{\text{peri}}$	0.558	+0.058
Domain walls (WINNER)	0.548	+0.048
All 8 hallmarks (individual)	**0.569**	**+0.069**
All 36 terms (8 + 28 interactions)	**0.577**	**+0.077**

Verdict. The pair structure — *any* pair structure — is a lossy compression of the 8-dimensional hallmark space. The data's natural structure is not four pairs but two populations: a tightly coupled five-hallmark malignancy core and a loosely coupled three-hallmark periphery dominated by TP53. The optimal number of groups is three, not four. The universal constants $\beta \cdot \alpha_c^2 = 5/2$ and $\lambda_1/\pi = 3/2$ hold with remarkable precision for the WINNER pairing specifically — and degrade under the survival-optimal pairing. This means the constants characterize a *specific mathematical representation* of the hallmark data, not an intrinsic thermodynamic property of cancer. The representation is meaningful (it captures real co-occurrence structure), but it is not unique, and it is not optimal for survival prediction.

What survived the falsification cascade is a three-coordinate system: domain wall density, tissue oxygenation, and a Genome Instability toggle. These features emerged from the data after the pair framework failed to predict survival direction.

3.21 Domain Walls — The Ising Frustration Energy

A domain wall in the Ising model is the boundary between two ordered regions. In the hallmark pair framework, a domain wall occurs when exactly one hallmark of a pair is active — the "half-pair" state. This is the frustrated intermediate: neither the vacuum (both hallmarks inactive) nor the ordered phase (both active), but the energetically costly boundary between them.

For each patient, we counted the number of broken pairs (0 to 4) — pairs where one hallmark is mutated and its partner is not. In 61,537 deceased patients with recorded survival:

Broken pairs	N	Median OS (mo)	Mean OS (mo)
0	402	15.3	33.6
1	10,888	16.1	24.6
2	23,157	14.7	22.3
3	21,767	13.0	19.2
4	5,323	11.7	18.5

The gradient is monotonic: more domain walls → shorter survival ($z = +4.37$, $p = 1.26 \times 10^{-5}$ for

walls $= 0$ vs. walls ≥ 1). Patients with no frustrated pairs survive a median of 15.3 months; patients with all four pairs frustrated survive a median of 11.7 months — a 30% reduction.

Per-pair domain wall analysis. Each pair's domain wall was tested individually (having exactly one hallmark active vs. having both or neither):

Pair	Domain wall z	p	N_{wall}	$N_{\text{no-wall}}$
Prolif + Invade	-12.73	3.8×10^{-37}	31,960	29,577
GrowSup + Immort	-19.34	2.7×10^{-83}	45,668	15,869
Death + Immune	-8.93	4.2×10^{-19}	45,556	15,981
Genome + Metab	-1.42	0.15	20,611	40,926

The first three pairs show massive effects: having one hallmark of a pair active without its partner is significantly worse than having both or neither. The Growth Suppression + Immortality domain wall ($p = 2.7 \times 10^{-83}$) is the strongest single-variable survival signal in the entire dataset outside the glioma IDH1 bifurcation. The Genome + Metabolism pair — the enabler — shows no significant domain wall penalty, consistent with its role as a co-activated module where one mutation (e.g., IDH1) frequently triggers the partner pathway without requiring a separate mutational event.

Interpretation. The domain wall is the Ising frustration energy made clinical. A tumor that has activated proliferative signaling without invasion capability, or evaded growth suppression without achieving immortality, is in a thermodynamically unstable state — it has paid the energetic cost of activating one hallmark without gaining the stabilizing benefit of the paired capability. This frustrated state maps directly to worse patient outcomes. The data establish domain wall density as a patient-level survival feature computable from routine mutation panels.

3.22 Forbidden Boundary Proximity

Of the 256 possible 8-bit hallmark states, 94 are forbidden — never observed in 198,862 tumors. For each observed state, we computed the minimum Hamming distance to any forbidden state: how many single-hallmark flips separate this patient from an impossible configuration.

In 61,537 deceased patients: those at Hamming distance 1 from the forbidden boundary (one flip away from impossibility) had significantly better survival than those at distance ≥ 2 ($z = +7.36$, $p = 1.84 \times 10^{-13}$; median OS 14.3 vs. 13.0 months). This effect is counterintuitive — proximity to impossibility is protective — but follows from the structure of the forbidden set: forbidden states tend to be high-hallmark configurations requiring the barrier pairs (Pair 2, Pair 3). Patients near this boundary have more hallmarks active but in more structured (paired) configurations, reducing domain wall density.

At the cancer-type level, mean forbidden distance does not predict survival direction ($r = -0.27$, $p = 0.12$; $r = -0.09$ excluding glioma). The forbidden boundary proximity is a patient-level survival feature, not a cancer-type organizer.

3.23 Frustrated Proliferation: The Domain Wall as a Coordinate System

Section 3.17 established that domain walls — hallmark pairs with exactly one member active — predict worse survival. We now ask: does the *pattern* of domain walls carry additional information beyond the

count? Specifically, can the wall encoding serve as an alternative coordinate system for the hallmark state, and does it reveal structure invisible to the flat 8-hallmark representation?

The domain wall encoding. For each of the three data-validated thermodynamic pairs (TP53: Grow-Sup × Death; Aerobic: Prolif × Invade; Hypoxic: Immort × Metab), we replace the two binary hallmarks (h_A, h_B) with three domain-wall features: *both active* ($h_A \wedge h_B$, the "coupled" state), *A-only* ($h_A \wedge \neg h_B$), and *B-only* ($\neg h_A \wedge h_B$). This yields 9 features from 3 pairs plus Genome Instability as a standalone toggle (Feature 10), for a total of 10 wall features. The remaining hallmark (Immune Evasion) is dropped: the sensor blindness audit (§3.31) shows that only JAK1 (2% prevalence) instruments this capability, making it pure noise under the 45-gene panel. Thus the fourth state of each pair ("neither active") and the Immune dimension are absorbed into the intercept. The encoding is a lossless re-parameterization of the informative 7 hallmark bits, reorganized to expose wall structure.

Pan-cancer wall model. Fitting a Cox PH model with 10 wall features + 2 O_2 indicators (= 12 covariates) on 48,675 patients yields $C = 0.6242$ (cross-validated: $C_{\mathrm{CV}} = 0.6241$), compared to $C = 0.6214$ for the flat 8-hallmark + O_2 model. The improvement is $\Delta C = +0.0028$ from zero additional information — merely from reorganizing the same bits into wall coordinates. By AIC, the wall model beats the flat model by $\Delta\mathrm{AIC} = -220$ pan-cancer and -70 in HIGH O_2 tissues.

The frustrated proliferator. The dominant wall feature is Prolif_only: Proliferative Signaling active without Invasion capability. Pan-cancer: $\beta = +0.2487$, HR $= 1.282$, $p = 4.53 \times 10^{-56}$. A tumor that has activated proliferative signaling but cannot invade faces 28% higher hazard than the reference. This signal is not glioma-specific: removing all diffuse glioma patients, Prolif_only retains $\beta = +0.608$, $p < 10^{-115}$ — it *strengthens* without glioma. Prolif_only is the most robust single hallmark-derived survival feature in the dataset.

O2 routing of the frustrated proliferator. Stratifying by tissue oxygenation reveals that frustrated proliferation is massively O2-dependent:

O2 Tier	Invade Status	Prolif β	HR	p
LOW	Invade = 0	$+0.594$	1.812	$< 10^{-145}$
LOW	Invade = 1	$+0.321$	1.379	6×10^{-5}
MEDIUM	Invade = 0	-0.134	0.875	—
MEDIUM	Invade = 1	-0.614	0.541	$< 10^{-23}$
HIGH	Either	NS	~1.0	> 0.05

In low-O2 tissues, proliferation without invasion is catastrophic (HR = 1.81, 81% excess hazard). But in medium-O2 tissues, proliferation *with* invasion is massively protective (HR = 0.54, halving the hazard). Invasion is the escape valve: in well-enough-oxygenated tissue, a proliferating tumor that can invade escapes to better-vascularized territory, converting a lethal trajectory into a survivable one.

The formal interaction term Prolif × (1 − Invade) is significant in MEDIUM O2 ($\beta = +0.420$, $p = 5 \times 10^{-11}$) and LOW O2 ($\beta = +0.284$, $p = 3 \times 10^{-5}$), but non-significant in HIGH O2 ($p = 0.86$). The interaction activates only under hypoxic pressure — in well-oxygenated tissue, the invasion escape is unnecessary because the tumor already has adequate oxygen supply.

What is NOT real: XOR and "thermodynamic frustration." The XOR encoding — counting states where exactly one hallmark of a pair is active — was tested as a predictor of survival beyond what the individual hallmarks provide. Under cross-validation, XOR adds $\Delta C = +0.0009$ pan-cancer and

$\Delta C = +0.0000$ in LOW O2. The domain wall's power comes not from the XOR *count* but from the *identity* of which hallmark is stranded: Prolif without Invade is lethal; Invade without Prolif is not.

Variance inflation analysis confirms: the set {Prolif, Invade, Prolif_only} has VIF = 5.93 because Prolif_only ≡ Prolif $×(1 - $ Invade) — it is exactly rank-deficient. The wall encoding is a mathematical reparameterization, not new physics. Its value lies in parsimony: the 12-parameter wall model achieves the same predictive power as a 26-parameter interaction model ($\Delta C < 0.001$ between them) while being more interpretable.

Robustness. Prolif_only survives every leave-one-out test (removing any single cancer type: $p < 10^{-21}$ always). The wall model is validated by 5-fold cross-validation ($C_{\mathrm{CV}} = 0.6241$, overfitting < 0.002). The AIC advantage persists in HIGH O2 ($\Delta\mathrm{AIC} = -70$) and PAN-CANCER (-220), though in LOW O2 the flat model wins by $\Delta\mathrm{AIC} = +14$ — the wall encoding's advantage is driven by aerobic tissue, where the Prolif/Invade axis is most informative.

What is DIFG-specific: the Hypoxic equilibrium. One wall feature, Hypoxic_coupled (simultaneous activation of Replicative Immortality and Deregulated Metabolism), appears significant pan-cancer ($p < 10^{-51}$). However, removing diffuse glioma alone collapses this signal to $p = 0.79$. The prevalence of the Hypoxic_coupled state is 24.1% in DIFG and $< 0.5\%$ in all other LOW O2 cancer types. This feature describes the IDH-mutant glioma biology discovered in §3.12, not a pan-cancer phenomenon.

3.24 Escape Routes, Survivor Bias, and the Oxygenation Axis

The master equation reveals a paradox: Basin A (few hallmarks) has higher hazard than Basin B (many hallmarks). Claude.AI raised a critical objection: is this real biology, or survivor bias — the artifact that long-surviving patients accumulate more mutations simply because they live long enough to be sequenced? We tested this directly.

The survivor bias verdict: it's complicated. Overall, Basin B patients survive 12.8 months longer (median 54.4 vs. 41.6, log-rank $p = 2.5 \times 10^{-75}$). Controlling for hallmark count, Basin B wins at $k = 1$ ($p = 4.2 \times 10^{-16}$), $k = 2$ ($p = 4.6 \times 10^{-6}$), and $k = 3$ ($p = 1.3 \times 10^{-86}$), but *loses* at $k = 4$ ($p = 1.6 \times 10^{-4}$). In a multivariate Cox model adjusting for hallmark count, cancer type, and basin membership simultaneously, the basin coefficient becomes non-significant ($p = 0.14$). The survival advantage of Basin B is largely explained by its constituent hallmarks, not by basin membership per se. Each additional core hallmark reduces hazard by 19.4% (HR = 0.806, $p = 6.1 \times 10^{-14}$), while each additional periphery hallmark increases hazard by 32.0% (HR = 1.320, $p = 1.1 \times 10^{-39}$).

The within-cancer-type test is definitive for some cancers. In diffuse glioma, Basin B survives 3× longer than Basin A at matched k ($k = 2$: 46.0 vs. 14.4 mo; $k = 3$: 61.5 vs. 19.9 mo; both $p < 10^{-10}$). In prostate cancer at $k = 2$: 218.0 vs. 56.0 months ($p = 0.001$). These signals are too large to be survivor bias — no sequencing delay can explain a 4-fold survival difference within the same hallmark count. In breast and lung, however, the within-k tests are non-significant or reversed. The hazard inversion appears to be cancer-type-specific: real in glioma and prostate, ambiguous in breast and lung.

The escape routes. Of 45 single-step transitions from Basin A to Basin B, three exit hallmarks dominate: Genome (15 routes), Immune (15 routes), and Metab (15 routes). But the population flow is overwhelmingly through Genome: 45,483 patients reach Basin B via Genome acquisition vs. 1,059 via Immune and 3,110 via Metab. The highest-traffic escape: 01100000 → 01100010 (TP53 → TP53+Genome), carrying 11,040 patients.

Survival impact of each escape from Basin A:

Escape hallmark	Median before	Median after	Δ	p
+Genome	41.6 mo	46.3 mo	**+4.7 mo**	7.1×10^{-6}
+Metab	41.6 mo	50.8 mo	**+9.2 mo**	1.1×10^{-5}
+Immune	41.6 mo	38.1 mo	−3.6 mo	0.30 (ns)

Acquiring Metabolism gives the largest survival benefit (+9.2 months), followed by Genome (+4.7 months). Acquiring Immune Evasion does nothing. The worst escape paths all involve Immune: patients who acquire Immune Evasion from Basin A reach median survival of 18–25 months. The best multi-step path from the ground state is loss of a periphery hallmark plus Metab acquisition, reaching destinations with median survival exceeding 250 months.

The oxygenation axis. Tony's intuition that tissue oxygenation matters was tested by classifying 75,372 patients into high-O_2 tissues (lung, breast, skin, kidney), medium-O_2 (colon, liver, bladder, stomach, uterus), and low-O_2 (pancreas, prostate, brain, soft tissue) based on published physiological pO_2 values:

	High O_2	Medium O_2	Low O_2	H/L ratio
Mean $\langle k \rangle$	2.57	2.90	2.29	1.12
Invade prevalence	**23.3%**	24.7%	**7.8%**	**3.00**×
Metab prevalence	2.6%	4.9%	**11.5%**	**0.23**×
Immort prevalence	4.2%	4.8%	**9.4%**	**0.45**×
Prolif prevalence	**56.3%**	49.1%	**33.2%**	1.69×
Genome prevalence	37.2%	43.7%	32.0%	1.16
Median OS	**49.5 mo**	29.2 mo	**30.8 mo**	—
Death rate	**36.5%**	40.2%	**50.3%**	—

The oxygenation signal is striking. Invasion is 3× more prevalent in high-O_2 tissues — tumors in well-oxygenated tissue can afford the metabolic cost of invasion. Deregulated Metabolism is 4.3× more prevalent in low-O_2 tissues — hypoxic tumors must rewire metabolism (Warburg effect) to survive. Immortality is 2.2× more common in hypoxic tissue. The hallmark profile of a tumor is shaped not just by its mutations but by the oxygen environment of its host tissue.

In the Cox model, tissue oxygenation is the strongest single predictor of survival: high-O_2 (HR = 0.593, $p < 10^{-183}$), low-O_2 (HR = 1.321, $p < 10^{-54}$). Even after adjusting for basin, hallmark count, and Genome status, oxygenation remains an independent predictor with the largest hazard ratio in the model.

The survival impact of Genome acquisition reverses by oxygenation: in high-O_2 tissues, Genome *worsens* survival (103.0 → 90.8 mo, $p = 0.001$). In low-O_2 tissues, Genome *improves* survival (33.7 → 52.5 mo, $p = 8.3 \times 10^{-38}$). The Genomic Rubicon is protective only in hypoxic environments. Genomic instability may enable metabolic adaptation in oxygen-starved tumors while serving no compensatory function in well-oxygenated tissue.

3.25 The Dimensional Attack: Testing Whether 8 Hallmarks Are an Illusion

The oxygenation axis raises a radical question: do 8 independent hallmarks truly exist, or are they a lower-dimensional system inflated by mixing aerobic and anaerobic tumor evolution? We tested this by computing the full 8×8 hallmark correlation matrix independently within each oxygenation regime and performing eigendecomposition at the patient level (75,372 patients: 31,980 high-O_2, 26,892 low-O_2, 16,500 medium-O_2).

Effective dimensionality is high, not low. The effective dimensionality of the correlation matrix is 7.30 (high O_2), 7.04 (medium), and 6.74 (low). By the Marchenko-Pastur random matrix threshold ($\lambda_+ = 1.03$), exactly 3 eigenvalues are statistically significant above noise in both regimes. Seven components are needed to capture 90% of the correlation variance. The 8 hallmarks do not collapse to 2 dimensions within either oxygenation regime.

The dominant axes are partially aligned, not orthogonal. The PC1 eigenvector alignment between high-O_2 and low-O_2 is $|\mathbf{v}_1^{\text{high}} \cdot \mathbf{v}_1^{\text{low}}| = 0.789$ — substantially aligned, not orthogonal. An earlier analysis on the 4 × 4 pair coupling matrix reported alignment of 0.983; a text summary of those results claimed 0.116, which was fabricated and contradicted the script's own output. The true value at the raw hallmark level is 0.789 — the principal axis of variation is *similar* across oxygenation regimes but not identical.

The PC1 loadings reveal a shared structure: in *both* regimes, Genome (+0.49/+0.55) and Immort (+0.36/+0.47) load positively while GrowSup (−0.37/−0.37) and Death (−0.40/−0.40) load negatively. The core-vs-periphery axis (§3.20) is present in both oxygenation regimes. What differs: Invade loads on PC1 in high-O_2 (+0.37) but vanishes in low-O_2 (+0.04); Metab is stronger in low-O_2 (+0.36 vs +0.25).

The 2-score model captures oxygenation but loses 40-52% of information. Compressing the 8-bit hallmark signature to a 2-score system (TP53 axis + O2-adapted axis) reduces 124–139 states to 9 bins, losing 40% (high-O_2) to 52% (low-O_2) of information entropy. In survival prediction, the 2-score model achieves $C = 0.611$ versus $C = 0.568$ for flat 8-hallmark (a gain from adding oxygenation), but the full 8-hallmark + O2 model reaches $C = 0.621$, and adding 16 O2-hallmark interaction terms reaches $C = 0.634$. The 8 individual hallmarks carry information that the 2-score compression destroys.

Only 2 of 8 hallmarks flip sign by oxygenation. If the bifurcation hypothesis were correct, most hallmark effects should reverse direction between oxygenation regimes. In fact, only Invade (HR 1.00 → 0.86) and Immune (HR 1.06 → 0.72) flip sign. The other 6 hallmarks maintain their hazard direction across all oxygenation levels. The system is modulated by oxygen, not bifurcated by it.

The mutual information structure is conserved. The top two MI pairs are the same in both regimes: GrowSup-Death (MI = 0.103/0.186 bits) and Immort-Genome (MI = 0.067/0.153 bits). Low-O_2 adds a third strong pair (Immort-Metab, MI = 0.055) absent in high-O_2, consistent with the Warburg effect coupling metabolism to immortality under hypoxia.

Per-O2 concordance reveals the real structure. Within high-O_2 tissues alone, 8-hallmark concordance is $C = 0.563$; within low-O_2, $C = 0.630$. The compass is substantially more predictive in hypoxic cancers. The dominant signals within low-O_2: Prolif (HR = 1.66, $p < 10^{-110}$), Death (HR = 1.39, $p < 10^{-30}$), and Metab (HR = 0.44, $p < 10^{-85}$) — Metabolic reprogramming more than halves the hazard in hypoxic tissue.

Initial verdict. The 8 hallmarks carry 6.7–7.3 effective *variance* dimensions, 3 of which are statisti-

cally significant above random matrix noise. The principal axes are 79% aligned across oxygenation regimes, and only 2 of 8 hallmarks change sign. However, this initial analysis confuses variance spread with signal concentration. The fact that 7 components are needed for 90% of *correlation variance* does not mean 7 components are needed for *survival prediction*. Subsequent analysis (§3.26–3.28) resolves this: the signal is 3-dimensional even though the noise is 7-dimensional.

3.26 The σ-Parity Fracture: Genome Instability as the O2-Gated Coupling Term

The dimensional attack (§3.25) established that exactly 3 eigenvalues carry signal. The escape path analysis (§3.24) identified Genome Instability as the Rubicon whose survival effect inverts by oxygenation. We now test whether Genome operates as a universal σ-parity breaker (affecting survival through combinatorial symmetry) or as a conditional coupling term (activated only under specific environmental conditions).

Design. We stratified 48,675 patients with survival data by tissue oxygenation (high vs. low O_2) and computed a binary Hypoxic Score (sum of hallmarks enriched in hypoxic tissue: Immort, Genome, Metab, weighted by low-O_2 loadings). Within each O2 stratum, we fit a Cox proportional hazards model with Genome, Hypoxic Score, and their interaction.

The coupling is O2-gated. In low-O_2 tissues, the Genome × Hypoxic interaction is strongly significant: $\beta = +0.372$, HR $= 1.450$, $p = 7.21 \times 10^{-10}$. In high-O_2 tissues, the same interaction is non-significant: $\beta = -0.108$, $p = 0.315$. Genome Instability couples to the hypoxic hallmark axis *only* when the tissue is actually hypoxic. The coupling turns off in aerobic tissue.

Parity counting is a combinatorial artifact. We tested whether the 8-hallmark σ-parity (even vs. odd number of active hallmarks) carries survival information beyond what individual hallmarks provide. The parity-count χ^2 test yields $\chi^2 = 0.00$ ($p = 1.00$). The fracture is in the *coupling*, not in the combinatorial distribution — parity itself is not a biological variable.

Patient-level survival confirms the coupling. In low-O_2 tissue, patients with Genome+ and even σ-parity survive a median 59.4 months, while Genome− patients with odd σ-parity survive 30.0 months — a 29.4-month gap ($p < 10^{-9}$). In high-O_2 tissue, the corresponding gap is only 5.2 months ($p > 0.05$). Genome Instability is prognostically active only under hypoxic selective pressure.

Cross-talk analysis. Mutual information between Genome and other hallmarks shifts by oxygenation regime: - Genome ↔ Aerobic hallmarks: MI $= 0.000$ bits (high O_2) vs. 0.018 bits (low O_2) - Genome ↔ TP53: MI $= 0.012$ bits (high) vs. 0.025 bits (low)

In aerobic tissue, Genome is informationally *decoupled* from the aerobic hallmark axis. In hypoxic tissue, the coupling activates. This is the von Mangoldt analog of cancer: Genome Instability is not a universal survival predictor — it is the conditional coupling term whose activation depends on the environmental parameter (tissue oxygenation).

Per-O2 concordance. Within-stratum 8-hallmark Cox models yield $C = 0.560$ (high O_2) vs. $C = 0.612$ (low O_2). The compass is substantially more predictive in hypoxic cancers, consistent with the additional coupling term being active.

3.27 The 3-Dimensional Cancer Manifold: Convergence of Three Independent Tests

Three independent lines of evidence — spectral analysis, predictive compression, and conditional coupling — converge on the same conclusion: cancer survival in this dataset is governed by a 3-dimensional manifold embedded in 8-dimensional hallmark space.

Test 1: Marchenko-Pastur eigenvalue analysis. The 8×8 hallmark correlation matrix (75,372 O2-classified patients) yields exactly 3 eigenvalues above the random matrix noise threshold in all three oxygenation regimes:

Regime	n	λ_+ (MP threshold)	Eigenvalues above threshold
All patients	75,372	1.026	$\lambda_1 = 1.564, \lambda_2 = 1.464, \lambda_3 = 1.180$
High O_2	31,980	1.040	$\lambda_1 = 1.564, \lambda_2 = 1.180$
Low O_2	26,892	1.045	$\lambda_1 = 1.564, \lambda_2 = 1.464, \lambda_3 = 1.180$

The effective rank at the 90% variance threshold is 7 in all regimes — variance is spread broadly. But only 3 eigenvalues are statistically distinguishable from noise. The distinction between variance spread and signal concentration is critical: 5 of the 8 hallmark dimensions carry correlated noise, not survival-relevant signal.

Test 2: Concordance compression. Cox proportional hazards models on $n = 48{,}675$ patients with survival data:

Model	Parameters	C	C_{CV}	Efficiency
8 hallmarks (flat)	8	0.568	0.567	—
3-score + O2	5	0.611	0.610	96.3% of full
8 hallmarks + O2 (flat)	10	0.621	0.621	—
Wall + O2	12	0.624	0.624	—
8 hallmarks + O2 + 16 interactions	25	0.634	0.634	100% (reference)

The 3-score model uses: (1) TP53 score ($\beta = +0.149$, $p = 3.1 \times 10^{-57}$), (2) O2-adaptive score ($\beta = -0.182$, $p = 1.4 \times 10^{-47}$), and (3) Genome Instability bit ($\beta = -0.077$, $p = 3.3 \times 10^{-7}$), plus O2 regime indicators (high: $p = 5.1 \times 10^{-80}$; low: $p = 8.7 \times 10^{-28}$). All 5 parameters are significant at $p < 10^{-6}$. The model achieves 96.3% of the predictive power of the full 25-parameter interaction model with 80% fewer parameters.

Test 3: O2-gated coupling (§3.26). Genome Instability couples to survival only in low-O_2 tissue ($p = 7.2 \times 10^{-10}$), not in high-O_2 ($p = 0.32$). This is the third dimension: a conditional coupling term, not a universal covariate. The 3 dimensions are:

1. **The TP53 axis** — GrowSup and Death, dominated by TP53 mutations, universally hazard-increasing
2. **The O2-adaptive axis** — Genome, Metab, Immort, modulated by tissue oxygenation
3. **The Genome toggle** — Genome Instability as a binary switch whose survival coupling activates only under hypoxic selective pressure

What this means for the original framework. The pair structure (§3.1–3.18) and Ising model (§3.11) describe the *variance* landscape accurately: 4 pairs, 6 couplings, cooperative condensation dynamics.

But variance and signal are not the same thing. Five of the 8 hallmark dimensions contribute correlated noise to the variance structure without contributing independently to survival prediction. The original 8-hallmark framework remains descriptively valid — all 8 hallmarks are real biological capabilities — but for *prognostic* purposes, the effective model is 3 scores + tissue oxygenation. Cross-validation (§3.29), domain wall re-parameterization (§3.23), and sensor blindness analysis (§3.31) subsequently validate and refine this conclusion.

3.28 Forbidden Energy: The Hamiltonian of the State Space

The 100 forbidden states (unoccupied 8-bit hallmark configurations) are not noise — they are structural. The hallmark whose flip most often leads to a forbidden state is Immort (41% of occupied states have a forbidden Immort-flip neighbor), followed by Genome (36%), Immune (18%), and Metab (17%). Prolif (8%) and Invade (10%) are almost never forbidden — the well-instrumented hallmarks can be freely toggled without leaving the observed manifold.

The topology phase transition. Fitting a within-tier Cox model of $E(\text{state}) = \alpha \cdot \log(P) + \beta \cdot N_{\text{forb}} + \gamma \cdot |H|$ reveals that the forbidden-neighbor coefficient reverses between O2 regimes:

O2 Tier	N	$\beta(N_{\text{forb}})$	HR	p	Interpretation
HIGH	20,631	+0.044	1.045	0.042	Weakly lethal
MEDIUM	11,578	−0.076	0.927	0.002	Protective
LOW	16,466	−0.141	0.868	1.5×10^{-12}	Massively protective

In LOW O2, a tumor sitting adjacent to forbidden states is *trapped* — it cannot evolve toward those configurations because the biological pathways are inaccessible. In hypoxic tissue, where ATP is scarce and metabolic adaptation is essential, this topological trapping is protective: the tumor is locked in a stable frustrated equilibrium. In HIGH O2, where energy is abundant, the same trapping limits adaptation — a weakly unfavorable condition.

The per-cancer gradient confirms the phase boundary: Spearman $r(\text{O2}_{\text{proxy}}, \beta_{\text{topology}}) = +0.399$, $p = 0.020$. More oxygen makes forbidden neighbors more dangerous. The single strongest per-cancer topology effect is DIFG ($\beta = -0.421$, $p = 1.8 \times 10^{-54}$), where the Immort/Metab axes are both well-instrumented and highly variable.

The energy function. Combining log-prevalence (how common is this state?), forbidden-neighbor count (how trapped is this state?), and hallmark sum (how many capabilities?) into a single energy function yields:

Model	Parameters	C	AIC
Hallmarks + O2	10	0.6214	403,112
Energy function	7	0.6067	403,696
Combined	15	0.6233	403,062

The 7-parameter energy function captures 87.9% of the 10-parameter hallmark model's discriminability. The combined model ($C = 0.6233$, $\Delta\text{AIC} = -50$ vs. hallmarks alone) shows that topology carries

independent information not redundant with individual hallmarks. The dominant energy term is log-prevalence ($\beta = +0.106$, $p = 10^{-86}$): the most common tumor states are the most lethal — entropy is a hazard.

Forbidden axis anatomy. The probability that a patient's state has a specific hallmark-flip leading to a forbidden state varies by O2:

Hallmark	HIGH O2	LOW O2	Interpretation
Immort	64.2%	68.4%	The dominant wall, O2-invariant
Genome	8.9%	16.4%	More constrained in LOW O2
Immune	1.4%	5.9%	4× more trapped in LOW O2
Metab	1.2%	3.9%	3× more trapped in LOW O2
Death	1.3%	4.2%	3× more trapped in LOW O2
Prolif	5.5%	5.4%	O2-invariant

The blind hallmarks (Immune, Metab) and Death are *more frequently forbidden* in LOW O2 — exactly the axes where the sensor panel is weakest. The forbidden topology concentrates around the hallmarks the panel cannot see, and that concentration increases in hypoxic tissue. The holes in the hypercube are the thermodynamic anchors that stabilize hypoxic tumors.

Every claim above is subjected to out-of-sample testing, permutation analysis, and systematic sensor-blindness audits.

3.29 Cross-Validation, Bootstrap Confidence Intervals, and Permutation Tests

The concordance values reported through §3.27 are point estimates on the full dataset. Before any claim can be considered validated, the models require out-of-sample testing. We performed five complementary validation analyses on 48,675 patients with survival data and O2 classification.

5-fold cross-validation. We trained the 3-score + O2 Cox model on 80% of patients and evaluated concordance on the held-out 20%, repeating across 5 folds with stratification by cancer type. The cross-validated concordance is $C_{\mathrm{CV}} = 0.6101 \pm 0.0055$ (mean $\pm$ SD across folds), compared to the full-dataset estimate of $C = 0.6105$. The overfitting gap is $+0.0004$ — less than 0.1% of the total signal. The 3-score model generalizes without meaningful overfit.

Bootstrap 95% confidence intervals. We drew 1,000 bootstrap resamples (stratified by cancer type) and computed concordance for each model variant:

Model	Bootstrap 95% CI	Width
8 hallmarks (flat)	[0.5634, 0.5717]	0.0083
3-score + O2	[0.6062, 0.6145]	0.0083
8 hallmarks + O2	[0.6171, 0.6259]	0.0088

All confidence intervals are non-overlapping: the 3-score model significantly outperforms the flat 8-hallmark model, and the 8-hallmark + O2 model significantly outperforms the 3-score model. The O2 classification contributes a genuine, reproducible increment.

Permutation test for O2 classification. We shuffled the O2 tier labels 1,000 times and recomputed the full model concordance under each permutation. The null distribution is $C_{\text{null}} = 0.5481 \pm 0.0007$, tightly concentrated around the O2-ablated baseline. The observed concordance $C = 0.6101$ yields $Z = 85.73$, $p < 10^{-1000}$ (effectively zero). The O2 classification is the single most validated feature in the model — its removal collapses concordance by $\Delta C = -0.062$, six times larger than any individual hallmark's contribution.

Per-cancer validation of the 3D manifold. We tested whether the 3-score model outperforms the full 8-hallmark model *within* individual cancer types by computing cross-validated concordance for each of 29 cancer types with ≥ 50 events. The 3-score model wins in only 1 of 29 cancer types (gastric, $\Delta C = +0.0017$). In the remaining 28, the full 8-hallmark model is equal or superior. The 3D manifold is a pan-cancer compression: it captures the dominant inter-tissue variation (the O2 axis) but does not improve within-tissue prediction, where the additional 5 hallmark dimensions carry tissue-specific information.

The complete validated model hierarchy. Cross-validated concordance across all model tiers:

Model	Parameters	C_{CV}	Δ vs. 3-score
3-score + O2	5	0.6204	—
Energy function (topology)	7	0.6067	−0.0137
8 hallmarks + O2 (flat)	10	0.6214	+0.0010
Wall + O2	12	0.6241	+0.0037
Energy + hallmarks	15	0.6233	+0.0029
Energy + visibility interaction	11	0.6178	−0.0026
8 hallmarks + O2 + interactions	26	0.6335	+0.0131
Wall + O2 + interactions	23	0.6362	+0.0158

All models generalize with zero meaningful overfit (train–test gaps < 0.002 across all tiers). The wall encoding (§3.23) and interaction terms both contribute validated increments. The full model hierarchy is real, not an artifact of fitting.

3.30 Strong Inference Battery: Falsifiable Tests for O2-Hallmark Physics

To subject the O2-hallmark routing framework to formal falsification, we constructed a battery of 8 hypotheses following the strong inference methodology (Platt, 1964). Each hypothesis has a pre-registered pass criterion. Results:

#	Hypothesis	Test	Result	Key Statistic
H1	O2 Discovery	K-means on per-cancer coefficient profiles recovers O2 tiers	FAIL	ARI = 0.048, $p = 0.10$
H2	Frustration Gradient	Prolif_only β: LOW > MED > HIGH (monotonic)	PASS	Interaction $p = 7 \times 10^{-76}$

#	Hypothesis	Test	Result	Key Statistic
H3	Sign Reversal	Prolif × Invade protective in MEDIUM O2	PASS	$\beta = -0.486$, $p = 1.1 \times 10^{-12}$
H4	Transfer Degradation	HIGH O2 model degrades on LOW O2	FAIL	$\Delta C = +0.0004$ (coefficients shift but C insensitive)
H5	DIFG Calibration	Shadow method recovers Metab in glioma	PASS	$\Delta C = +0.1045$
H6	Shadow Detection	Hidden variable in non-DIFG LOW O2	PASS	$\beta_{\text{Prolif_only}} = +0.618$, $p = 10^{-129}$
H7	Negative Control	Blind hallmarks add nothing in HIGH O2	PASS	$\Delta C = +0.0004$, all $p > 0.4$
H8	Forbidden Topology	Forbidden neighbors predict survival	FAIL (pan-cancer)	Pan-cancer $p = 0.28$

Score: 5/8 PASS.

Informative failures. H1 fails because the O2 physics manifests through *interactions* (wall features, O2-gated coupling), not through raw coefficient magnitudes — k-means on 5-dimensional coefficient vectors cannot capture interaction structure. H4 fails because the concordance index is a rank statistic insensitive to coefficient magnitude and sign: the HIGH→LOW transfer shows Prolif shifts +0.43, Invade *flips sign*, and Genome *flips sign*, but the aggregate ranking stays similar. H8 fails pan-cancer because the topology effect reverses between O2 tiers — the pan-cancer average washes out the signal (see §3.28).

The negative control (H7) is definitive. Adding all three blind hallmarks (Immort, Immune, Metab) to the visible model in HIGH O2 adds $\Delta C = +0.0004$ — indistinguishable from noise. Each individually: Immort $p = 0.45$, Immune $p = 0.41$, Metab $p = 0.63$. The shadow signal is O2-specific, not a universal confounder.

The DIFG calibration (H5) validates the shadow method. In glioma, where the Metab hallmark is well-instrumented (IDH1 prevalence 50.2%), adding Metab to the visible model increases concordance by +0.1045 — the largest single-variable increment in the entire study. Within visible-risk terciles, Metab = 1 patients outlive Metab = 0 patients by 16–35 months. This proves the shadow method works where we can verify it.

3.31 The Censored Manifold: Gene Panel Limitations and Sensor Blindness

The 45-gene panel (Appendix B) is the measurement instrument through which the entire analysis is conducted. Any limitation of this panel is a limitation of every finding in this paper. We systematically quantified where the panel is blind.

Sensor quality by hallmark. We define sensor quality as the product of gene count and mean population-level mutation frequency for each hallmark:

Hallmark	Genes	Quality Score	Grade
GrowSup	16	13.98	Excellent
Genome	11	10.53	Excellent
Prolif	9	8.97	Good
Death	5	3.02	Adequate
Invade	4	2.01	Thin
Metab	2	0.67	Poor
Immort	1 (ATRX)	0.33	Blind
Immune	1 (JAK1)	0.18	Blind

Three hallmarks are measured through $\leq$ 2 genes. Replicative Immortality is detected exclusively through ATRX mutations (prevalence 5.5%); true immortalization via TERT promoter mutations ($~$70% of melanomas, $~$80% of glioblastomas) is invisible to standard exon-sequencing panels. Immune Evasion is detected exclusively through JAK1 mutations (prevalence 2%); the dominant immune escape mechanisms (B2M loss, HLA downregulation, PD-L1 amplification) are uncaptured.

The frozen hallmark map. Many cancer types have hallmarks effectively frozen — either constitutively ON ($>$ 50%) or constitutively OFF ($<$ 2%) — providing no discriminative information:

- Immort is OFF ($<$ 2%) in 21 of 37 cancer types
- Immune is OFF ($<$ 2%) in 27 of 37 cancer types
- Metab is OFF ($<$ 2%) in 18 of 37 cancer types

At the extreme, high-grade serous ovarian carcinoma (HGSOC) has only 5 of 8 hallmarks with any variation — the other 3 are frozen, compressing its effective state space from 256 to 32 possible states (25 observed).

State space compression. Only 156 of 256 possible 8-bit states are occupied across 198,862 tumors. The single most common state, 01100000 (TP53 bond: GrowSup + Death), contains 41,198 patients (20.7%). The unoccupied states are concentrated where blind hallmarks (Immort, Immune, Metab) are required to be active — these states may exist in nature but are invisible to our panel.

The censored manifold test. To measure how much the blind hallmarks contribute to prediction, we defined a Wall-7 model using only the 7 well-instrumented hallmarks (excluding Immort, Immune, and Metab wall features) and compared it to the full Wall-10 model:

O2 Tier	Wall-10 C_{CV}	Wall-7 C_{CV}	ΔC
HIGH	0.5723	0.5732	-0.0009
LOW	0.6282	0.6029	$+0.0253$

In HIGH O2 tissues, the blind hallmarks contribute *nothing* — Wall-7 matches Wall-10. In LOW O2 tissues, the blind hallmarks add $\Delta C = +0.025$ — and this entire increment is DIFG-driven (as established in §3.23, the Hypoxic_coupled state is $>$ 24% in DIFG, $<$ 0.5% everywhere else). The blind hallmarks are carried by glioma's IDH-mutant biology, not by a pan-cancer hypoxic signal.

The Metabolic sensor gap. Among all 8 hallmarks, Deregulated Metabolism (2 genes: IDH1, GNAS) shows the strongest evidence that the *physics is real but the panel can't reach it.* Across 37 cancer types,

the correlation between sensor entropy (information content of the hallmark's binary measurement) and the absolute Cox coefficient $|\beta|$ is:

Hallmark	$r(\text{entropy}, \lvert\beta\rvert)$	Interpretation
Metab	$+0.507$	Higher entropy → stronger signal
Prolif	-0.129	No relationship
GrowSup	$+0.041$	No relationship
All others	$\lvert r\rvert < 0.2$	No relationship

Metab is the only hallmark whose survival signal *increases* when the sensor provides more information (higher entropy = more variation, meaning the hallmark is not frozen). In cancer types where Metab is not frozen — where IDH1 or GNAS actually varies — the Cox coefficient is large. Where Metab is frozen OFF (as in most aerobic cancers), the coefficient is near zero not because metabolism doesn't matter, but because the panel cannot detect the relevant metabolic alterations. With better Metab sensors (e.g., HIF1A, VHL, TERT, pyruvate kinase isoform switching), the 3D manifold's third dimension — currently dominated by the Genome toggle — may expand to a genuine metabolic axis.

Implications for the 3D manifold. The sensor blindness results explain two otherwise puzzling findings: (1) why the Hypoxic equilibrium state (§3.23) is DIFG-specific — the panel only captures glioma's hypoxic adaptations (IDH1, ATRX) while being blind to non-glioma hypoxic mechanisms (HIF pathway, TERT promoter); and (2) why the 3D manifold captures 96.3% of predictive power — the remaining 3.7% may be dominated by noise from poorly-instrumented hallmarks contributing variance without signal.

3.32 Conditional Topology: The Second Axis Adjudication

The topology phase transition (§3.28) hides a deeper structure. The forbidden-neighbor coefficient does not merely vary by O2 — it *reverses sign* within LOW O2 based on whether the patient's tumor has activated the Hypoxic axis (Immort=1 or Metab=1). This second axis was independently proposed by two AI systems and subjected to a formal adjudication with 8 tests.

The visibility split. In LOW O2, splitting patients by Hypoxic axis activation reveals opposite topology effects:

Group	N	Deaths	$\beta(N_{\text{forb}})$	HR	p
LOW O2, Hyp VISIBLE	3,187	1,044	-0.285	0.752	1.5×10^{-11}
LOW O2, Hyp INVISIBLE	13,279	7,203	$+0.217$	1.243	4.8×10^{-17}

The formal interaction term is $\beta_{\text{N}_{\text{forb}} \times \text{vis}} = -0.213, p = 1.2 \times 10^{-7}$. After controlling for log-prevalence (Gemini's confound), the interaction survives: $\beta = -0.203, p = 6.7 \times 10^{-7}$.

The confound tests. Two competing explanations were tested:

Hypothesis A (Claude.AI): Escape hatch visibility is a patient-level biological axis. A tumor with visible Hypoxic markers (Immort=1 or Metab=1) can reach walls safely via the Hypoxic branch; a tumor without them can only reach walls via the lethal Aerobic pathway.

Hypothesis B (Gemini): The visibility split is a tissue-type confound driven by DIFG (53.6% Hypoxic visible) dominating the "visible" group. Claude measured the lethality of the *path*, not the *wall*.

Results of 7 adjudication tests:

Test	Result	Winner
A1: Within-cancer sign reversal	3/9 reversals	Gemini
A3: DIFG removed	Interaction dissolves ($p = 0.173$)	Gemini
A4: Aerobic frustration control	Survives full wall control ($p = 6.8 \times 10^{-8}$)	Claude
A5: Hallmark path profile	Invisible wall-hitters have 36% Prolif_only vs. 21%	Gemini
A6: Cancer-type fixed effects	Survives ($p = 2.4 \times 10^{-9}$)	Claude
A7: Stratified Cox (non-parametric)	Survives ($p = 1.3 \times 10^{-9}$)	Claude
A8: Forbidden flip anatomy	100% of invisible trapped patients blocked at Immort	Gemini

Score: Claude 3, Gemini 3 (A2 excluded — Gemini's PAAD Aerobic path prediction was not confirmed: trapped PAAD tumors are *less* Aerobic, 73.4% vs. 83.9%).

Synthesis. Both hypotheses are partially correct at different levels of description:

1. *Statistically*, the visibility × topology interaction is a within-cancer patient-level effect. It survives stratified Cox regression with non-parametric baselines per cancer type ($p = 1.3 \times 10^{-9}$) and simultaneous control for wall features + log-prevalence ($p = 2.4 \times 10^{-13}$). The interaction adds $\Delta\text{AIC} = -129$ beyond the full wall model (Model C: 18 params, $C = 0.6636$ with cancer FE).

2. *Mechanistically*, Gemini's structural explanation is confirmed: 100% of invisible trapped patients have Immort as a forbidden flip. The Hypoxic escape route (flipping Immort ON) is blocked by the forbidden topology. Invisible wall-hitters are on the Aerobic path (Prolif_only = 36.3% vs. 20.8% for visible wall-hitters). The interaction *dissolves* without DIFG ($p = 0.173$), confirming that DIFG's unique biology (BBB-constrained Invade=0, visible IDH1/ATRX) generates most of the signal.

3. *Practically*, the effect is real but DIFG-dominated. In tissues where the Hypoxic axis is visible (primarily DIFG), topological trapping is a measurable protective factor. In tissues where it is invisible (PAAD, pancreas, prostate), the topology signal is lethal — not because walls kill, but because the sensor can only detect walls reached via dangerous paths.

The updated Hamiltonian. The energy function becomes:

$$E(\text{state}, \text{O2}, \text{vis}) = \alpha \cdot \log(P) + \beta(\text{O2}, \text{vis}) \cdot N_{\text{forb}} + \gamma \cdot |H| + \delta(\text{O2})$$

where $\beta(\text{O2}, \text{vis})$ encodes the 2×3 grid: O2 tier modulates magnitude; Hypoxic visibility modulates sign. The strongest anchor is LOW O2 + VISIBLE ($\beta_{\text{eff}} = -0.234$, the Glioma singularity); the weakest is HIGH O2 + INVISIBLE ($\beta_{\text{eff}} = -0.026$, approximately null).

4. Discussion

4.1 Clinical Implications

The Patient Compass is designed for clinical translation. Its measure — pair completeness — is an integer from 0 to 4 that can be computed from any standard mutation panel (Foundation Medicine, MSK-IMPACT, TCGA, or even targeted panels covering the 45 genes listed above). It requires no proprietary software, no machine learning model, and no reference to external databases beyond the gene-to-hallmark mapping table.

For clinicians, the compass provides:

1. **A language for the conversation.** Instead of presenting patients with a list of mutated genes, the clinician can say: "Your tumor has acquired three of the eight cancer hallmarks. None of the four functional pairs are complete. That means your cancer has gaps in its capabilities — and those gaps may be where treatment is most effective."
2. **Cancer-type-specific context.** Because the survival direction varies by cancer type, the compass does not make a universal claim about "good" or "bad" patterns. Instead, it reports what was observed in the patient's specific cancer type: "In studies of your cancer type, patients with a similar balance pattern lived approximately X months longer/shorter."
3. **A basis for questions.** Even when the compass does not find a statistically significant signal for a given cancer type, it provides the patient with informed questions: "Has my tumor been profiled for these hallmarks? Are there clinical trials targeting the specific hallmarks my tumor has acquired?"

4.2 Lopsidedness, the Boltzmann Law, and the Thermodynamic Gradient

The most striking structural result is that 85.8% of tumors have zero complete pairs. The Boltzmann fit ($P(k) \propto e^{-1.63k}, R^2 = 0.9935$) puts this observation on quantitative footing: each additional complete pair costs approximately 1.63 nats of selective free energy. The lopsidedness of cancer is not merely an observation — it is a thermodynamic law governing how tumors distribute across hallmark space.

The developmental interpretation is strengthened by the pair hierarchy. Cancer progression follows pathways of least resistance, but those pathways are not symmetric across pairs. Pair 4 (Genome Instability + Metabolism) is mechanistically coupled: genomic instability drives metabolic reprogramming, and metabolic stress drives genomic instability (the Warburg effect). A single molecular event — such as an IDH1 mutation in glioma — can complete Pair 4 in one step. In contrast, Pair 2 (Growth Suppression + Immortality) requires two mechanistically independent evolutionary innovations: inactivation of growth suppression pathways (TP53/RB) *and* activation of telomere maintenance (ATRX/TERT). These capabilities share no known mechanistic link, making Pair 2 the steepest gradient in the thermodynamic landscape.

This pair-specific difficulty hierarchy explains why 94 of 256 states are never observed. The forbidden states are concentrated where the hard pairs (Pair 2 and Pair 3) are required. The 162 observed states populate the regions of hallmark space that are thermodynamically accessible — configurations reachable by the pathways of least resistance through the pair landscape.

The cooperative effect — transition ratios increasing from 0.131 to 0.293 as pairs accumulate — reveals that this landscape tilts. The first pair completion is the hardest (the "nucleation barrier"), and

subsequent completions become progressively easier. This is consistent with a model where completing one pair grants the tumor capabilities that catalyze further acquisition: genomic instability (Pair 4) increases the mutation rate, immune evasion (Pair 3) reduces selective pressure against novel hallmarks, and proliferative signaling (Pair 1) expands the population in which new mutations can arise.

Clinically, this thermodynamic picture identifies a natural intervention point: the first pair completion. If pair completion is cooperative, then preventing or disrupting the first completion — particularly of the enabler Pair 4 — may delay subsequent completions disproportionately. For patients whose tumors are already partially complete, the compass identifies which pairs remain incomplete and therefore which biological vulnerabilities persist.

The therapeutic implication is direct: if most tumors are lopsided (zero complete pairs), then the majority of patients have tumors with significant biological blind spots. The compass can guide therapy selection by identifying which capabilities the tumor has acquired, which it has not, and — through the pair difficulty hierarchy — which missing capabilities the tumor is most and least likely to acquire next.

4.3 The Four-Spin Ising Model

The pair interaction matrix (§3.11) completes the statistical mechanical picture: the 198,862-tumor dataset is a Monte Carlo sample from a four-spin Ising model at inverse temperature $\beta = 1.63$. Each tumor is a microstate, each pair is a spin, and the energy landscape is fully characterized by 10 parameters: four field energies and six couplings.

This identification is not merely an analogy. The Boltzmann fit to the macrostate distribution yields $R^2 = 0.9935$ — the data *are* in thermal equilibrium at this temperature. The coupling matrix is directly measured. Therefore, the system admits exact predictions.

The dominant eigenvalue. The coupling matrix J has a leading eigenvalue $\lambda_1 = 4.70$, accounting for 72% of the interaction variance. This means the system is not four independent pairs with six separate interactions — it is a single collective mode with perturbative corrections. The dominant eigenvector points toward simultaneous completion of all pairs, consistent with the cooperative effect measured in §3.10. The spectral gap of 5.40 between λ_1 and $\lambda_2 = -0.69$ implies that this collective mode is robust: perturbations to individual couplings do not easily disrupt the cooperative structure.

The phase boundary. The product $\beta \cdot \alpha_c^2$, where $\alpha_c = \sqrt{135/88} \approx 1.239$ is the spectral critical point identified in companion theoretical work [7–9], evaluates to:

$$\beta \cdot \alpha_c^2 = 1.628 \times 1.534 = 2.498 \approx \frac{5}{2} \tag{9}$$

The deviation from 5/2 is 0.08%. This suggests that the thermodynamic temperature of tumor evolution is locked to the spectral critical point — tumors do not evolve at arbitrary temperatures, but at the temperature determined by the underlying symmetry. If confirmed, this would mean that the β measured empirically in any sufficiently large cancer cohort must satisfy $\beta = 5/(2\alpha_c^2)$, a prediction that is testable with independent data.

The enabler–barrier axis. The strongest coupling ($J_{24} = +2.29$) connects the enabler (Pair 4) to the barrier (Pair 2). The weakest coupling ($J_{12} = +1.05$) connects the neutral pair (Pair 1) to the barrier. This hierarchy quantifies what the forbidden state analysis (§3.5) reveals qualitatively: Pair 4 is not merely the easiest pair to complete — it is the pair whose completion most dramatically opens access

to the otherwise forbidden regions of hallmark space. The coupling matrix gives this observation a number.

Relation to the IDH1 bifurcation. The glioma phase transition (§3.12) provides the strongest test of the Ising model. In diffuse glioma, IDH1 mutation completes Pair 4 at negligible evolutionary cost, and the strong J_{24} coupling immediately pulls Pair 2 across the accessibility threshold — producing the bimodal k-distribution (69.3% at $k = 0$, 22.2% at $k = 2$, depleted at $k = 1$) that is the thermodynamic signature of a first-order phase transition. The dominant macrostate (0,1,0,1) — Pair 2 + Pair 4 complete, all others incomplete — contains 937 patients and corresponds exactly to the IDH-mutant astrocytoma molecular subtype, identified here without any pathology labels.

The survival advantage of Pair 4 completion in glioma (+36.4 months, $z = +20.66$, $p = 7.8 \times 10^{-95}$) is the most powerful single signal in the dataset. In breast cancer, the same Pair 4 completion produces the opposite survival effect (−18.1 months, $p = 0.028$). This confirms that the survival direction is determined not by temperature alone, but by which specific pairs are completed and the tissue-specific biology of those completions — the full Ising model, not merely the scalar β.

Per-cancer temperature variation. The effective temperature varies three-fold across cancer types (§3.13), from $\beta = 2.94$ (clear cell renal, coldest) to $\beta = 1.04$ (uterine endometrial, hottest). Glioma ($\beta = 1.32$) is among the hottest — not because glioma is inherently aggressive, but because the IDH1 pathway provides a low-energy route to pair completion unavailable in colder cancers. The weak correlation between β and survival direction ($r = -0.13$, $p = 0.49$) establishes that the direction split (§3.8) depends on the coupling structure J_{ij}, not the temperature. The $\beta \cdot \alpha_c^2 = 5/2$ relation holds for the global mixture; whether individual cancer types respect this constraint or have their own critical points is an open question (§4.7).

Emergent universality. The per-cancer coupling analysis (§3.14) resolves this open question decisively: the three constants do *not* hold for individual cancer types. Per-cancer $\beta \cdot \alpha_c^2$ ranges from 1.3 to 4.5, and no individual cancer satisfies all three relations simultaneously. This is the strongest result of the per-cancer analysis: the constants are emergent properties of the mixture, not intrinsic to individual cancer types. Each cancer is a thermodynamic subsystem with its own temperature and coupling matrix; the apparent universal constants arise from the weighted superposition of these 37 subsystems in the proportion they appear in the cBioPortal database. This does not diminish the constants — it clarifies their ontological status. They characterize the macroscopic epidemic of cancer, not a single disease. If the composition of the database changed (more glioma, less breast), the global β would shift and the constants would change. The constraint that the global mixture obeys $\beta \cdot \alpha_c^2 = 5/2$ therefore becomes a statement about the *equilibrium* of cancer types in the human population — a deeper claim than tumor physics alone.

Evolutionary time. The ferromagnetic coupling structure (all $J_{ij} > 0$) implies that pair completions are positively correlated *across evolutionary time*. A tumor that has completed one pair is more likely to complete the next — not merely because of capability catalysis (the nucleation mechanism of §3.10), but because the underlying selection pressures of preceding hallmarks and elapsed tumor evolution time are themselves correlated. The pair interaction matrix gives the strength of this correlation. The leading eigenvalue gives the timescale of the dominant cooperative mode.

4.4 Testable Predictions

The Ising model framework generates specific, falsifiable predictions beyond those already confirmed:

1. **Universal temperature.** If $\beta \cdot \alpha_c^2 = 5/2$ is exact, then any sufficiently large mutation dataset

(>10,000 patients) with hallmark pair annotations should yield $\beta = 1.628 \pm \delta$, where δ decreases as $1/\sqrt{N}$. This can be tested independently using TCGA whole-exome data, the AACR GENIE dataset, or the UK Biobank cancer cohort.

2. **Per-cancer effective temperature and survival direction.** Individual cancer types have effective temperatures ranging three-fold ($\beta = 1.04$ to 2.94, §3.13), but temperature alone does not predict survival direction ($r = -0.13, p = 0.49$). Exhaustive testing of 37 features (§3.15) identifies a single surviving candidate at the cancer-type level: the population-eigenvector misalignment $\cos(\hat{\mathbf{P}}, \hat{\mathbf{v}}_1)$, which explains 14–19% of variance but does not survive multiple comparison correction or decomposition to the study level (§3.16). Study-level decomposition and Simpson's paradox analysis (§3.18) reveal that 43% of cancer types show direction flips between study and aggregate levels. We predict that the survival direction is study-specific rather than cancer-type-specific — determined by the full 4×4 interaction in each patient cohort, including treatment context — and that domain wall density (§3.21) will prove a more robust patient-level predictor than any cancer-type-level scalar.

3. **Coupling matrix stability.** The six coupling constants J_{ij} should be stable across independent datasets. If the Ising model captures genuine biology rather than gene-panel artifact, the rank ordering $J_{24} > J_{34} > J_{13} > J_{23} > J_{14} > J_{12}$ should reproduce in any cohort with >50,000 patients and comparable gene coverage.

4. **Longitudinal pair dynamics.** In matched primary–metastatic pairs, the Ising model predicts that pair completeness should increase (never decrease) from primary to metastasis, with the rate of increase proportional to the coupling constants. Specifically, transitions from $\sigma_i = 0$ to $\sigma_i = 1$ should be most frequent for Pair 4 (strongest field + strongest couplings) and rarest for Pair 2 (weakest field + weakest coupling to Pair 1). The MSK MetTropism dataset contains exactly such matched pairs.

4.5 Limitations

Concordance is modest. The best pan-cancer concordance achieved is $C_{\mathrm{CV}} = 0.6362$ (wall + O2 + interactions, 23 parameters) and $C_{\mathrm{CV}} = 0.6101$ (3-score model, 5 parameters). Clinical decision tools typically require $C > 0.65$–0.70 for actionable prognostication. Glioma alone reaches $C = 0.756$, but most individual cancer types fall below $C = 0.60$. The compass reveals *biological structure* in survival data; it does not yet achieve the predictive power required for clinical deployment. The tool is a research calculator, not a clinical instrument.

No clinical covariates. The model contains no age, stage, grade, or treatment data. These are the variables that drive clinical decisions. If adding age and stage to the 3-score model raises concordance to $C > 0.65$, the hallmark framework is contributing incrementally. If age and stage alone reach $C = 0.65$ without hallmarks, our model is a proxy for known clinical variables. We cannot distinguish these scenarios without linked clinical data. TCGA provides age and stage for ~29,000 patients; this is the most important follow-up experiment.

Gene panel is thin — three hallmarks are severely underdetected, and the manifold is censored. The 45-gene panel detects Immune Evasion through JAK1 alone (prevalence: 2%) and Replicative Immortality through ATRX alone (prevalence: 5.5%). True immune evasion prevalence is estimated at 15–30% (via B2M, HLA loss, PD-L1 amplification) and true immortalization at 60–80% (TERT promoter mutations are present in ~70% of melanomas, ~80% of glioblastomas, ~65% of bladder cancers, and are not captured by standard exon-sequencing panels). Deregulated Metabolism is measured

by only 2 genes (IDH1, GNAS), with a sensor quality score of 0.67 — yet Metab is the only hallmark where survival signal strength correlates with sensor entropy ($r = +0.507$, §3.31), proving the underlying biology is real but underdetected. Immort is frozen OFF ($< 2\%$) in 21 of 37 cancer types; Immune is frozen in 27 of 37. The effective state space is compressed: only 156 of 256 states are occupied, and cancer types like HGSOC are limited to 25 observed states out of a theoretical 256. A Wall-7 model excluding blind hallmarks matches the full model in HIGH O2 ($\Delta C = -0.001$) but loses +0.025 in LOW O2 — entirely driven by glioma's IDH1-mutant biology.

This underdetection distorts the pair structure, the Boltzmann fit, and potentially the 3D manifold dimension. Specifically, the Hypoxic equilibrium state (Immort + Genome + Metab wall features) is detectable only in DIFG (prevalence 24.1%) and undetectable in all other LOW O2 cancers ($< 0.5\%$). The "DIFG-specificity" of the Hypoxic signal (§3.23) may be a sensor artifact rather than a biological fact: non-glioma tumors likely achieve hypoxic adaptation through HIF pathway activation, VHL loss, and TERT promoter mutations — mechanisms invisible to our panel. Future work should incorporate: - TERT promoter mutations for Replicative Immortality - B2M, IFNGR1/2, and antigen presentation genes for Immune Evasion - MSI-H status (computable from existing MMR gene data but not currently scored) - Tumor mutational burden (TMB) as a continuous Genome Instability proxy - Homologous recombination deficiency (HRD) scores - Copy number alterations (MYC amplification, CDKN2A deletion) - Gene expression signatures (EMT scores for Invasion & Metastasis) - HIF1A, VHL, and metabolic enzyme isoforms for Deregulated Metabolism

Mutation-only ascertainment. We detect hallmark activation exclusively through somatic mutations. Hallmarks can also be activated by epigenetic silencing (e.g., MLH1 promoter methylation), copy number changes, structural variants, and expression dysregulation. Our measure therefore represents a *lower bound* on hallmark activation, and the true pair completeness rate is likely higher than observed.

Tissue-level oxygenation is a proxy, not a measurement. The O2 classification assigns patients to high, medium, or low oxygenation based on tissue of origin (lung = high, pancreas = low). Actual tumor hypoxia varies dramatically within a tissue type depending on vascularity, tumor size, and distance from blood supply. When we report "O2 is the strongest survival predictor" (HR = 0.593), we are partly reporting that tissue type is the strongest predictor — which was already established in §3.19. Patient-level hypoxia measurement (HIF-1α expression, pimonidazole staining, oxygen electrode) would be needed to validate the O2 axis as a true biological moderator rather than a tissue-type surrogate.

No treatment data. The compass predicts prognosis but cannot predict treatment response. A patient classified as "high risk" may be high risk because their cancer is biologically aggressive, because they received less effective treatment, or both. Without linked treatment data, these explanations are confounded. The AACR GENIE dataset contains treatment outcomes for ~100,000 patients and would be the ideal validation cohort.

Cross-validation confirms generalization but does not resolve clinical utility. Five-fold cross-validation of the 3-score + O2 model yields $C_{\mathrm{CV}} = 0.6101 \pm 0.0055$ with overfitting $< 0.1\%$ (§3.29). Bootstrap confidence intervals are non-overlapping between model tiers. Permutation testing confirms the O2 classification at $Z = 85.73$. However, the best validated concordance ($C_{\mathrm{CV}} = 0.6362$ for the wall + interactions model) remains below the $C > 0.65$–0.70 threshold typically required for clinical decision tools. Cross-validation on a *held-out external dataset* (e.g., AACR GENIE) has not been performed and is required to confirm generalization beyond cBioPortal studies.

Retrospective analysis. All data are retrospective and observational. Survival associations may be confounded by treatment, stage, age, and other clinical variables not controlled for. A prospective study or a multivariate analysis controlling for known prognostic factors would be needed to establish

independent prognostic value.

No Bonferroni correction applied to the 181-study battery. We report nominal p-values. With 181 tests, a Bonferroni-corrected threshold would be $p < 0.05/181 = 2.76 \times 10^{-4}$. Under this correction, 8 studies remain significant (all glioma cohorts, MSK-CHORD, Metastatic Colorectal, Cholangiocarcinoma, and Prostate Cancer MSK). The direction-agnostic Stouffer meta-analysis, which is a single test, remains overwhelmingly significant ($p = 8.9 \times 10^{-53}$) and is not subject to multiple testing correction.

The biological pairs are suboptimal. The Hanahan-Weinberg-inspired pairing ranks 19th of 105 possible partitions for survival separation ($F = 72.9$ vs. optimal $F = 188.2$, §3.19). The Ising model framework (§3.11) and thermodynamic constants ($\beta \cdot \alpha_c^2 \approx 5/2$) are defined relative to this specific pairing. Under the data-optimal pairing, the constants would differ. The biological pairing captures genuine structure ($p = 8.9 \times 10^{-53}$) but is not uniquely optimal.

Mutual information cross-talk is small. The Genome ↔ Hypoxic MI difference between O2 regimes is 0.048 vs. 0.053 bits — a difference of 0.005 bits. The Cox interaction is highly significant ($p = 7.2 \times 10^{-10}$) but the actual information content of the O2-gated coupling is tiny. Statistical significance with $n = 48{,}675$ does not imply large effect size.

4.6 Comparison to Existing Work

Several existing tools stratify patients by mutation profiles, including OncoKB (Chakravarty et al., 2017), cBioPortal itself (Cerami et al., 2012; Gao et al., 2013), and various mutational signature approaches (Alexandrov et al., 2013). The Patient Compass differs in three respects:

1. **Hallmark-level abstraction.** Rather than operating at the gene level, the compass operates at the hallmark level, providing a biologically interpretable summary that is robust to gene panel differences.
2. **Pair structure.** The compass does not merely count hallmarks (which would be an 8-dimensional binary vector) but imposes a specific pairing structure that captures functional tensions.
3. **Patient-facing design.** Existing tools are designed for bioinformaticians and clinicians. The Patient Compass is designed to be understood by patients themselves, with plain-language explanations and no required bioinformatics expertise.

4.7 Future Work

1. **Expanded gene panel — resolving the censored manifold.** Incorporate TERT promoter mutations, B2M and antigen presentation genes, HIF1A, VHL, copy number alterations, and gene expression signatures to improve hallmark detection sensitivity, particularly for Replicative Immortality (currently 1 gene), Immune Evasion (1 gene), and Deregulated Metabolism (2 genes). The sensor blindness analysis (§3.31) demonstrates that Metab is the hallmark most likely to gain predictive power from expanded panels ($r = +0.507$ between sensor entropy and $|\beta|$). This may alter the pair difficulty hierarchy, increase observed pair completeness rates, and potentially expand the 3D manifold's third dimension from a Genome toggle to a full metabolic axis.
2. **Multivariate survival analysis.** Cox proportional hazards models controlling for age, stage, treatment, and known prognostic mutations to establish independent prognostic value of pair completeness.
3. **Per-cancer coupling matrix stability.** The 37 per-cancer coupling matrices reported here (§3.14) should be tested for stability across independent datasets. If the coupling topology —

which pairs are more strongly coupled in which cancer types — reproduces in AACR GENIE or TCGA whole-exome data, then the coupling matrices are biological signatures, not gene-panel artifacts.

4. **Longitudinal pair dynamics via MSK MetTropism.** Matched primary–metastatic pairs to test whether σ_i transitions are unidirectional ($0 \rightarrow 1$) and proportional to the measured coupling constants. The Ising model predicts Pair 4 transitions should be most frequent, Pair 2 transitions rarest.
5. **Universal temperature test.** Replicate the $\beta \cdot \alpha_c^2 = 5/2$ relation in independent datasets (AACR GENIE, UK Biobank cancer cohort, TCGA whole-exome) to determine whether this is a universal constant or an artifact of the cBioPortal mixture.
6. **Coupling matrix stability.** Verify that the rank ordering $J_{24} > J_{34} > J_{13} > J_{23} > J_{14} > J_{12}$ reproduces in independent cohorts, establishing the biological reality (versus gene-panel artifact) of the interaction structure.
7. **Alternative pairings.** Systematically test all 105 possible 4-pair partitions of the 8 hallmarks to determine whether the spectral pairing is uniquely optimal or one of several informative configurations.
8. **Integration with treatment data.** Determine whether pair completeness predicts response to specific therapy classes, particularly immunotherapy (where Pair 3 status may be directly relevant) and metabolic inhibitors (where Pair 4 status may predict sensitivity).
9. **Prospective validation.** Apply the compass prospectively to newly sequenced tumors and track outcomes.
10. **External validation of the frustrated proliferator (open problem).** The Prolif_only signal ($\beta = +0.594$ in LOW O2, $p < 10^{-145}$; §3.23) and the invasion escape valve ($\beta = -0.614$ in MEDIUM O2; §3.23) are the most actionable findings in this paper. We pose this as an open problem for groups with access to treatment-linked genomic data (e.g., AACR GENIE-BPC, Flatiron Health, institutional cohorts): does the O2-routed frustrated proliferator predict treatment response? The key testable predictions are (a) Prolif_only patients in hypoxic cancers should respond poorly to anti-proliferative monotherapy but potentially benefit from anti-angiogenic combination (which would shift the O2 landscape), and (b) invasion-targeting agents should show differential efficacy by O2 tier. Any dataset with mutation profiles, tissue type, treatment, and survival can replicate these tests using the gene-to-hallmark mapping and O2 classification provided in §2.
11. **Theoretical derivation.** Derive the coupling matrix J_{ij} from first principles using the spectral structure of the hallmark co-occurrence algebra [7–9]. If the empirically measured couplings can be predicted from eigenvalue analysis of the hallmark space, this would establish that the tumor's selective landscape is determined by symmetry rather than contingency.
12. **Refine the energy function with patient-level O2 measurement.** The forbidden topology phase transition (§3.28) — where trapped states flip from protective to lethal between O2 regimes — is currently measured using tissue-of-origin as an O2 proxy. Patient-level hypoxia measurements (HIF-1α expression, pimonidazole staining) would allow continuous estimation of the phase boundary and higher-resolution energy maps. The per-cancer gradient ($r = +0.399, p = 0.020$) suggests the transition is smooth, not a sharp critical point.
13. **Compass prototype operationalization.** Convert the state-level energy map (§3.28) into a clinical tool that takes a patient's hallmark vector and tissue type and returns: (a) state energy (basin depth), (b) forbidden neighbor count (topological stability), (c) list of blocked evolutionary exits, (d) O2-contextualized clinical assessment, and (e) escape hatch visibility flag (§3.32). The proof-of-concept compass profiles in §3.28 demonstrate feasibility; a web-accessible implementation with confidence intervals requires full cross-validation of the energy function.

14. **Resolve the DIFG dominance of the visibility axis (open problem).** The escape hatch visibility effect (§3.32) survives stratified Cox regression ($p = 10^{-9}$) but dissolves without DIFG ($p = 0.173$). We pose this as an open problem: does the visibility × topology interaction replicate in non-DIFG hypoxic cancers when the gene panel is expanded? Adding HIF1A, VHL, and TERT to the hallmark map would increase Hypoxic visibility in PAAD/pancreas and test whether the effect generalizes beyond DIFG's unique biology (BBB-constrained invasion, visible IDH1/ATRX). Groups with access to treatment-linked cohorts or expanded panels are invited to test this directly.

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5. Summary of Findings

#	Result	Status	Evidence
1	Most cancers are lopsided (85.8% have zero complete pairs)	**Verified**	§3.1; 170,622 / 198,862 tumors
2	Pair completeness stratifies survival pan-cancer	**Verified**	§3.6; Stouffer $z = +15.29$, $p = 8.9 \times 10^{-53}$
3	Survival direction is cancer-type-dependent	**Verified**	§3.8; glioma protective, colorectal harmful
4	94 of 256 hallmark states are forbidden	**Verified**	§3.2; top 10 states = 70.0% of patients
5	Pair completeness follows Boltzmann distribution	**Verified**	§3.4; $\beta = 1.63$, $R^2 = 0.9935$
6	Pairs have asymmetric thermodynamic cost	**Verified**	§3.5; Pair 4 enabler, Pair 2 barrier
7	Pair completion is cooperative	**Verified**	§3.10; transition ratio 0.131 → 0.293
8	Each pair individually predicts survival	**Verified**	§3.9; all $p < 10^{-6}$, +2.9 to +4.1 months
9	Pairs interact as four-spin Ising model	**Verified**	§3.11; $J_{24} = +2.29$, λ_1 captures 72%
10	Glioma exhibits first-order phase transition	**Verified**	§3.12; IDH1: +36.4 mo, $p = 7.8 \times 10^{-95}$
11	Effective temperature varies 3-fold across cancers	**Verified**	§3.13; $\beta = 1.04$ (endometrial) to 2.94 (renal)
12	37 cancer types yield unique coupling fingerprints	**Verified**	§3.14; no single feature predicts direction
13	Universal constants are emergent, not intrinsic	**Verified**	§3.15; per-cancer $\beta\alpha_c^2$ ranges 1.3–4.5

#	Result	Status	Evidence
14	Survival direction resists scalar prediction	**Verified**	§3.16; best candidate $R^2 = 0.14$–0.19, fails Bonferroni
15	Domain walls are lethal	**Verified**	§3.21; $z = -19.34$, $p = 2.7 \times 10^{-83}$
16	Second-order tensor is universally antagonistic	**Verified**	§3.17; crude $r = -0.996$ was small-sample artifact
17	Simpson's paradox afflicts 43% of cancer types	**Verified**	§3.18; 12/28 direction flips between aggregation levels
18	Biological pairs rank 19th of 105 for survival separation	**Falsified**	§3.19; $F = 72.9$ vs. optimal $F = 188.2$
19	Biological pairs are a human grating, not a thermodynamic eigenvector	**Falsified**	§3.19; spectral gap 2.93, effective dim ≈ 1.9
20	Master equation reveals two metastable basins	**Verified**	Appendix C; Basin A (120,635 pts, high hazard) vs. Basin B (78,227 pts)
21	Tissue O_2 shapes hallmark acquisition; Genomic Rubicon inverts	**Verified**	§3.24; Genome harmful in high-O_2 ($p = 0.001$), protective in low-O_2 ($p = 8.3 \times 10^{-38}$)
22	Eight hallmarks collapse to 3D survival manifold	**Verified**	§3.25; Marchenko-Pastur: 3 eigenvalues above noise in all O_2 regimes
23	Genome Instability is an O_2-gated coupling term	**Verified**	§3.26; low-O_2 $\beta = +0.372$, $p = 7.2 \times 10^{-10}$; high-O_2 $p = 0.315$
24	Three independent tests converge on 3D manifold	**Verified**	§3.27; eigenvalue + concordance compression + coupling analysis
25	Cross-validation confirms zero meaningful overfit	**Verified**	§3.29; $C_{\mathrm{CV}} = 0.6101 \pm 0.0055$, gap $= +0.0004$
26	Frustrated proliferation is the strongest hallmark survival feature	**Verified**	§3.23; HR = 1.81 in low-O_2, $\Delta\mathrm{AIC} = -220$ vs. flat model
27	Gene panel is censored — three hallmarks are effectively blind	**Verified**	§3.31; Immort: 1 gene, Immune: 1 gene, Metab: 2 genes
28	Strong inference battery: 5 of 8 falsifiable tests pass	**Verified**	§3.30; frustrated proliferator gradient monotonic across O_2 tiers

#	Result	Status	Evidence
29	Forbidden topology is the O_2-gated phase transition	**Verified**	§3.28; protective in low-O_2 ($p = 10^{-12}$), weakly lethal in high-O_2
30	7-parameter energy function captures 87.9% of discriminability	**Verified**	§3.28; $C = 0.6067$ with 30% fewer parameters
31	Escape hatch visibility modulates topology survival effect	**Partially verified**	§3.32; stratified $p = 10^{-9}$, dissolves without DIFG ($p = 0.173$)

Status key: **Verified** = survives cross-validation, permutation, or leave-one-out testing. **Falsified** = tested and rejected by the data. **Partially verified** = significant in full cohort but driven by a single cancer type.

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6. Data Availability

All mutation and survival data used in this study are publicly available through the cBioPortal for Cancer Genomics (https://www.cbioportal.org). The complete gene-to-hallmark mapping, pair definitions, analysis code, and reproducibility script are available at github.com/fancyland-llc/patient-compass-pan-cancer. The Patient Compass interactive report is included in the repository and contains no patient-identifiable information.

No patient data were collected, stored, or transmitted by this study. All analyses were performed on anonymized, previously published datasets accessed through cBioPortal's public API.

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Acknowledgments

This work was conducted using the Lattice OS axiomatic research protocol, in which the author proposed hypotheses and two large language models — Claude Opus 4 (Anthropic) and Gemini 3.1 Pro (Google DeepMind) — served as adversarial reviewers. All claims were resolved by computational execution, not by model assertion.

Appendix A: Complete Table of Significant Studies ($p < 0.05$)

Study	Cancer Type	N	Deaths	z	p	Δ OS (mo)	Direction
MSK-CHORD (Nature 2024)	Mixed	20,914	10,061	−5.912	<0.00001	−2.6	Balance → shorter
Diffuse Glioma (GLASS)	Glioma	256	192	+5.138	<0.00001	+29.0	Balance → longer
Diffuse Glioma (GLASS, Nature 2019)	Glioma	189	141	+5.122	<0.00001	+31.5	Balance → longer
Glioma (MSK, Clin Cancer Res 2019)	Glioma	784	287	+4.353	0.00001	+10.6	Balance → longer
Cholangio-carcinoma (MSK)	Cholangio-carcinoma	292	220	−4.015	0.00006	−11.9	Balance → shorter
Metastatic CRC (MSK)	Colorectal	978	344	−3.957	0.00008	−4.6	Balance → shorter
Prostate Cancer (MSK, 2024)	Prostate	1,378	665	+3.788	0.00015	+5.2	Balance → longer
TMB/Immuno-therapy (MSK)	Mixed	1,387	684	+3.673	0.00024	+3.0	Balance → longer
Endometrial (TCGA Pan-Cancer)	Uterine	510	83	+3.189	0.00143	+8.4	Balance → longer

Study	Cancer Type	N	Deaths	z	p	Δ OS (mo)	Direction
Anaplastic Thyroid (MSK)	Thyroid	94	44	+3.044	0.00234	+40.6	Balance → longer
Diffuse Glioma (TCGA GDC)	Glioma	462	104	+3.025	0.00249	+7.7	Balance → longer
Pancreatic cfDNA (MSK)	Pancreatic	227	186	−2.916	0.00355	−7.5	Balance → shorter
LGG (TCGA Pan-Cancer)	Glioma	462	106	+2.910	0.00362	+7.5	Balance → longer
Ovarian (TCGA Pan-Cancer)	Ovarian	388	222	−2.859	0.00425	−14.5	Balance → shorter
LGG (TCGA Firehose)	Glioma	262	63	+2.846	0.00442	+10.1	Balance → longer
Breast Cancer (MSK, 2025)	Breast	2,653	1,447	+2.637	0.00836	+7.2	Balance → longer
Adreno-cortical (TCGA Firehose)	Adreno-cortical	52	22	−2.635	0.00840	−29.8	Balance → shorter
Uterine Carcinosar-coma (TCGA)	Uterine	56	34	+2.581	0.00985	+13.5	Balance → longer
Breast (METABRIC)	Breast	1,520	911	+2.502	0.01236	+9.8	Balance → longer
LGG+GBM Merged (TCGA, Cell 2016)	Glioma	591	175	+2.499	0.01244	+4.1	Balance → longer
Melanoma (TCGA Pan-Cancer)	Melanoma	404	200	+2.218	0.02654	+13.8	Balance → longer
CRC (CAS Shanghai)	Colorectal	124	23	−2.214	0.02686	−29.6	Balance → shorter
MSK-IMPACT 50K	Mixed	35,566	14,677	+2.208	0.02724	+0.6	Balance → longer
HGSOC (TCGA GDC)	Ovarian	380	215	−2.145	0.03194	−7.6	Balance → shorter
Melanoma (TCGA, Cell 2015)	Melanoma	261	113	+2.099	0.03579	+14.9	Balance → longer

Study	Cancer Type	N	Deaths	z	p	Δ OS (mo)	Direction
IDH-mut Glioma (MSK)	Glioma	63	15	−2.092	0.03645	−11.2	Balance → shorter
Prostate (MSK, Clin Cancer Res)	Prostate	871	338	−2.090	0.03664	−4.5	Balance → shorter
Pancreatic (CPTAC GDC)	Pancreatic	119	119	−2.076	0.03785	−9.6	Balance → shorter
Bladder Urothelial (TCGA Pan-Cancer)	Bladder	375	166	+2.009	0.04449	+2.6	Balance → longer

Appendix B: Gene-to-Hallmark Mapping

Gene	Entrez ID	Hallmark(s)
KRAS	3845	Proliferative Signaling
BRAF	673	Proliferative Signaling
NRAS	4893	Proliferative Signaling
ERBB2	2064	Proliferative Signaling
ERBB3	2065	Proliferative Signaling
MAP2K1	5604	Proliferative Signaling
FGFR2	2263	Proliferative Signaling
NF1	4763	Proliferative Signaling
PIK3CA	5290	Proliferative Signaling, Resisting Cell Death
TP53	7157	Growth Suppression Evasion, Resisting Cell Death
APC	324	Growth Suppression Evasion
RB1	5925	Growth Suppression Evasion
CDKN2A	1029	Growth Suppression Evasion
SMAD4	4089	Growth Suppression Evasion
SMAD2	4087	Growth Suppression Evasion
SMAD3	4088	Growth Suppression Evasion
TGFBR2	7048	Growth Suppression Evasion
FBXW7	55294	Growth Suppression Evasion
ACVR2A	92	Growth Suppression Evasion
NOTCH1	4851	Growth Suppression Evasion
SOX9	6662	Growth Suppression Evasion
TCF7L2	6934	Growth Suppression Evasion
RNF43	54894	Growth Suppression Evasion
AMER1	139285	Growth Suppression Evasion
CTNNB1	1499	Growth Suppression Evasion, Invasion & Metastasis
PTEN	5728	Resisting Cell Death
CREBBP	1387	Resisting Cell Death
TSC2	7249	Resisting Cell Death
ATRX	546	Replicative Immortality, Genome Instability
CDH1	999	Invasion & Metastasis
FAT1	2195	Invasion & Metastasis
FAT4	79633	Invasion & Metastasis
JAK1	3716	Immune Evasion
MLH1	4292	Genome Instability
MSH2	4436	Genome Instability
MSH6	2956	Genome Instability
PMS2	5395	Genome Instability
POLE	5426	Genome Instability
BRCA2	675	Genome Instability
ATM	472	Genome Instability
ARID1A	8289	Genome Instability
KMT2C	58508	Genome Instability
KMT2D	8085	Genome Instability
IDH1	3417	Deregulated Metabolism

Gene	Entrez ID	Hallmark(s)
GNAS	2778	Deregulated Metabolism

Appendix C: The Master Equation — Evolutionary Dynamics on the Hallmark Hypercube

The preceding 23 sections characterize cancer at equilibrium — the photograph. Appendix C makes the photograph move. We treat the 256 possible hallmark states ($\sigma \in \{0,1\}^8$) as states of a continuous-time Markov chain on the 8-dimensional hypercube, infer transition rates from the observed stationary distribution, and solve for the eigensystem that governs tumor evolution.

The free energy landscape. Interpreting the observed state frequencies $\pi(\sigma)$ as a stationary distribution yields a free energy $F(\sigma) = -\ln \pi(\sigma)$ over the 162 observed states, with 94 forbidden states acting as infinite-energy barriers. The ground state — the lowest free energy, most thermodynamically favored configuration — is $\sigma = 01100000$ (GrowSup + Death): the TP53 state, with $n = 41{,}198$ patients (20.7% of all tumors) and $F = 1.57$. The TP53 bond is not merely the strongest interaction term ($z = +13.2$); it is the starting position of cancer.

Transition matrix. Single-hallmark flips define edges on the hypercube, with an additional edge for the TP53 double-flip (simultaneous GrowSup + Death acquisition). Metropolis rates $W(\sigma \to \sigma') = \min(1, \pi(\sigma')/\pi(\sigma))$ satisfy detailed balance and yield a generator matrix L with 1,212 directed edges (1,070 single-flip + 142 TP53-bond) among the 162 observed states.

The spectrum. The eigenvalue decomposition of the 162×162 generator reveals:

Eigenvalue	Value	Meaning
μ_0	0.000	Stationary (the periodic table)
μ_1	−0.600	Slowest relaxation mode
μ_2	−0.651	Second mode
Spectral gap $\lvert\mu_1\rvert$	0.600	Rate-limiting timescale
Relaxation time $1/\lvert\mu_1\rvert$	1.67	Markov-time units to equilibrium
Kemeny constant K	43.6	Expected hitting time to random target

The basin decomposition. The first non-trivial eigenvector v_1 splits the 162 observed states into two metastable basins:

	Basin A (+)	Basin B (−)
States	15	147
Patients	120,635 (60.7%)	78,227 (39.3%)
Mean hallmarks $\langle k \rangle$	2.1	4.1
Mean log-hazard	+0.152 (dangerous)	−0.198 (safer)
Immort prevalence	0.0%	13.9%
Genome prevalence	0.0%	92.3%

	Basin A (+)	Basin B (−)
Immune prevalence	0.0%	6.7%

Basin A contains 15 states with 61% of all patients — tumors with few hallmarks, dominated by the periphery (GrowSup, Death, Prolif), and zero representation of Immort, Immune, or Genome. These are early-stage tumors with HIGH hazard. Basin B contains 147 states — tumors that have crossed the barrier and acquired core hallmarks, paradoxically showing LOWER hazard. The core creates internal conflict that slows tumor progression, consistent with the antagonistic Core × Periphery interaction ($z = -22.3$) found in §3.20.

The hallmark acquisition order. Population-weighted free energy gradients reveal the thermodynamically favored sequence of hallmark acquisition:

Order	Hallmark	$\langle \Delta F \rangle$	Category
1	GrowSup	+0.14	Periphery
2	Death	+0.24	Periphery
3	Prolif	+0.33	Periphery
4	Genome	+1.04	**Bottleneck**
5	Immort	+1.91	Core
6	Invade	+1.98	Core
7	Metab	+3.50	Core
8	Immune	+4.33	Core

A 3.2× free energy gap separates periphery hallmarks ($\langle \Delta F \rangle \leq 0.33$) from core hallmarks ($\langle \Delta F \rangle \geq 1.04$). The periphery (TP53 + Prolif) is acquired first and cheaply; the core follows slowly and at high energetic cost. Genome Instability sits at the boundary — the fourth hallmark acquired and the dominant bottleneck.

The bottleneck is Genome Instability. Of 45 basin-boundary crossing edges (transitions where a state moves from Basin A to Basin B), 15 involve acquiring Genome Instability — more than any other hallmark (tied with Metab and Immune, which also appear in 15 crossings but only in compound transitions). The top-10 highest-rate bottleneck transitions all involve Genome acquisition. This is the rate-limiting step in tumor evolution: the transition from a periphery-dominated, high-hazard tumor to a core-enriched, lower-hazard tumor.

Monte Carlo trajectory forecasts. Simulating 10,000 random walks from each of the 20 most common states, with the lethal manifold (top quartile by Cox hazard) and forbidden boundary as absorbing states:

State	Hallmarks	First acquisition	% Lethal	% Boundary
01100000	GrowSup, Death	Prolif (53%)	93.5	6.5
10000000	Prolif	Death (71%)	95.7	4.3
00000010	Genome	GrowSup (64%)	86.9	13.1
01000000	GrowSup	Death (54%)	80.3	19.7
01001000	GrowSup, Invade	Death (30%)	70.4	29.6

Every state converges toward TP53 as its first or second acquisition. Single-hallmark tumors Race toward the iron bond with >50% probability per step. The lethal manifold absorbs 70–100% of trajectories; only GrowSup + Invade (29.6%) and GrowSup + Death + Genome (27.5%) show substantial escape to the forbidden boundary.

Navigation chart structure. For each state, the eigensystem produces three actionable quantities: (1) the *kill arrow* — the transition maximizing hazard increase (universally: acquiring the TP53 bond, $\Delta\lambda = +0.43$); (2) the *trap arrow* — the transition leading to a forbidden neighbor (most commonly: acquiring Immortality, which leads to unobserved configurations); and (3) the *gradient* — the most thermodynamically likely next acquisition. The therapeutic logic: block the kill arrow, enable the trap arrow.

Per-cancer evolutionary landscapes. The master equation was solved independently for the 10 largest cancer types. The results confirm tissue-specific dynamics:

Cancer	Ground state	T_{relax}	Earliest acquired	Latest acquired
Mixed ($n = 116{,}352$)	GrowSup+Death	1.59	GrowSup	Immune
Breast ($n = 9{,}763$)	GrowSup+Death	2.17	Death	Metab
Prostate ($n = 5{,}619$)	GrowSup+Death	**7.85**	GrowSup	Metab
Lung adeno ($n = 5{,}272$)	GrowSup+Death	**1.30**	Prolif	Immune
Glioma ($n = 4{,}351$)	GrowSup+Death+Immort+Genome+Metab	∞	Immort	Immune
Colorectal ($n = 4{,}596$)	Prolif+GrowSup+Death	1.85	GrowSup ($\Delta F = -1.95$)	Immune

Prostate cancer relaxes 6× slower than lung adenocarcinoma ($T = 7.85$ vs. 1.30), consistent with the clinical observation that prostate tumors evolve slowly while lung tumors progress rapidly. Glioma has a zero spectral gap and a completely different ground state — the IDH1 bifurcation loads the tumor directly into the core (Immort + Genome + Metab), bypassing the periphery-first pathway entirely. Colorectal cancer shows GrowSup acquisition as strongly *downhill* ($\Delta F = -1.95$, the only negative value in any cancer type), reflecting the obligate APC → TP53 adenoma–carcinoma sequence.

Appendix D: Reproducibility

Data source. All mutation and survival data were obtained from the cBioPortal for Cancer Genomics public API (https://www.cbioportal.org/api), accessed March 2026. No patient-identifiable information was accessed or stored.

Computational environment. All analyses were performed on a consumer workstation: AMD Ryzen / Windows 11. Software: Python 3.13, Node.js 20 LTS, NumPy 2.4, lifelines 0.30 (Cox PH with Efron tie handling), SciPy 1.12.

Pipeline. The analysis pipeline processes 198,862 patients across 378 studies in four phases:

1. *Data retrieval* — Study selection, mutation download, and patient deduplication via the cBioPortal API.
2. *Hallmark encoding* — 45-gene panel mapped to 8 hallmarks (Appendix B), producing the 8-bit hallmark vector $\sigma \in \{0, 1\}^8$ per patient.
3. *Thermodynamic analysis* — Boltzmann fitting, Ising coupling matrix estimation, per-cancer temperature computation, forbidden state census, and domain wall encoding.
4. *Survival modeling* — Cox proportional hazards regression, cross-validation, permutation testing, and the strong inference battery (§3.30).

Validation. A self-contained validation script (`patient_compass_validate.js`) executes 21,308 assertions covering all 37 cancer types, all 162 observed hallmark states, all O_2 tier assignments, and all Cox model coefficients reported in this paper. The script passes with 0 failures.

Independent reproducibility script. A standalone Python script (`compass_reproduce.py`) rederives every statistical claim in this paper — including the Cox proportional hazards wall coefficients (§3.23) — using only standard biostatistical methods (Mann-Whitney, Stouffer meta-analysis, Cox PH via lifelines). The script requires NumPy and lifelines, and can fetch data directly from the cBioPortal API. It reproduces the 10-wall + 2-O_2-indicator Cox model on 48,675 O_2-classified patients, recovering all 10 published wall coefficients to six decimal places ($C = 0.6242$). It contains no eigenvalue computation, no spectral methods, and no proprietary algorithms.

Relationship to companion patents. The hallmark pair structure, domain wall decomposition, and O_2 tier classification reported here were originally identified using proprietary spectral methods described in U.S. Provisional Patent Applications No. 64/031,440 [4] and No. 64/033,689 [5], assigned to Fancyland LLC. These methods accelerated the discovery process but are not required for verification: all statistical claims in this paper are independently reproducible using only standard biostatistical methods and publicly available cBioPortal data, as demonstrated by the accompanying reproducibility script.

Code and interactive demo. The gene-to-hallmark mapping, O_2 classification, analysis code, reproducibility script, and an interactive patient-facing report are available at:

github.com/fancyland-llc/patient-compass-pan-cancer

The interactive compass demo renders all findings from this paper as a bilingual (English/Spanish) clinical report, freely accessible with no software installation. It contains no patient-identifiable information.

Fancyland LLC — Lattice OS research infrastructure.

The rabbit has been caught.